Chapter2 Entity-Relationship (ER) Model

CHAPTER OVERVIEW

Entity-Relationship (ER) modeling serves as the conceptual design phase in database development, acting as a crucial bridge between real-world requirements and software implementation. By translating unstructured business descriptions into precise visual schemas, ER diagrams establish a shared blueprint for business stakeholders and database engineers before any physical tables or SQL code are created. This chapter covers the database design lifecycle, the core components of conceptual data modeling (entities, attributes, and relationships), Chen notation visual symbols, key and structural constraints, and variations in relationship degrees and cardinalities.

🎯 Learning Outcomes

At the end of this chapter, you will be able to:

1. **Explain the DB Design Lifecycle & Conceptual Schema Design:** Trace the four phases of database design (Requirements Engineering, Conceptual Design, Logical Design, and Physical Design) and define the purpose of an ER diagram as an implementation-independent model.
2. **Differentiate ER Diagram Notations:** Compare visual notation dialects—including academic **Chen notation**, formal **Min-Max structural constraints**, and industry-standard **Crow's Foot notation**—and identify their primary use cases.
3. **Identify & Classify Entity Types:** Differentiate between **strong entities** (having independent existence and primary key attributes) and **weak entities** (relying on an owner entity and a partial key via identifying relationships).
4. **Classify & Map Attribute Types:** Analyze data properties and represent them in Chen notation as **simple**, **composite**, **single-valued**, **multi-valued**, **derived**, **stored**, **key**, or **partial key** attributes.
5. **Define & Model Relationship Properties:** Construct and interpret relationships across varying degrees (**unary/recursive**, **binary**, **ternary**), applying attributes to relationships and enforcing structural constraints such as **cardinality ratios** (1:1, 1:N, M:N) and **participation rules** (total vs. partial).

> The conceptual schema is the key to database design; all else is implementation detail.
>
> — Peter Chen, 1976

1 Scenario: "What Do You See in This Restaurant?"

🤔 Warmup Question

Imagine we are sitting at a table in a busy restaurant, and you look at me and ask, "*Can you describe what you see?*"

Click to expand

Naturally, I wouldn't start reciting code or listing software specifications. I would describe the scene in plain text:

- "*I see customers sitting at tables.*"
- "*There's a server taking an order from a table.*"
- "*The kitchen is cooking dishes listed on the menu.*"
- "*When the meal is done, the server brings a bill to the customer.*"

This human ability to observe a real-world environment and describe its key elements is the exact starting point of **Data Modeling**.

However, computer systems and database developers cannot build a software system directly from plain text descriptions. Plain English can be ambiguous, vague, or open to misinterpretation. To bridge the gap between human language and actual database implementation, we need a structured, standardized way to express these plain text requirements. This is where **Conceptual Modeling** and **Entity-Relationship Diagrams (ERDs)** come in. They take those casual real-world observations and translate them into a clear visual blueprint of **Entities** (`Customer` , `Order` , `Menu_Item`), **Attributes** (`Customer_Name` , `Price`), and **Relationships** (*A Customer places an Order*).

Describing a scene as a human is fundamentally the same mental process as **Data Modeling** in computer science.

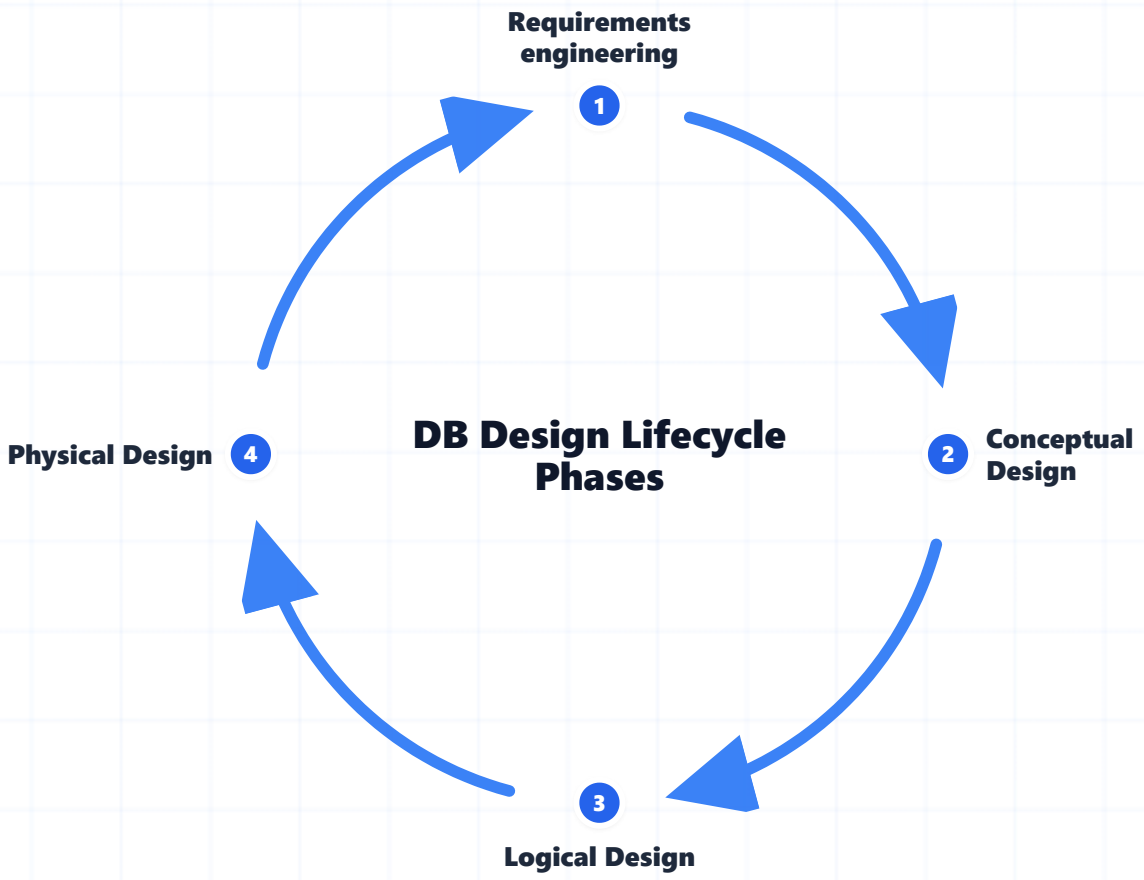
> 📖 DEFINITION 1: **Data Model**
>
> Concepts for describing data objects and their relationships (meta model).

> 📝 EXAMPLE 1: **Real-World Example: E-Commerce Store**
>
> Imagine building an online store. Before creating databases or writing code, you define the core concepts of the business:
>
> - **Entities (The Main Objects):**
> - `Customer` : The person making a purchase.
> - `Order` : The transaction placed by a customer.
> - `Product` : The item being sold.
> - **Attributes (Key Characteristics):**
> - `Customer` : Name, Email Address.
> - `Order` : Order Date, Total Amount.
> - `Product` : Product Name, Price.

- **Relationships & Cardinalities (How They Connect):**
  - A `Customer` **places** an `Order` (One customer can place **many** orders, but an order belongs to **one** customer).
  - An `Order` **contains** a `Product` (One order can contain **many** products, and a product can appear in **many** orders).

2 DB Design Lifecycle Phases

No database should be designed from thin air. A well-established **design lifecycle** guides the process from understanding what users need to deploying a production system.



1 Requirements engineering

- Collect and analyze data and application requirements
- **Key Questions**: What data must be stored? What

2 Conceptual Design

- Model data semantics and st ructure, independent of logi cal data model

3 Logical Design

- Model data with implementa tion primitives of concrete d ata model.

queries must be answered?
- **Deliverable:** Data requirements document.

- **Key Questions**: What entities and relationships exist in the domain?
- **Deliverable:** ER diagram.

- **Key Questions**: How do entities translate to tables? Are they normalized?
- **Deliverable:** Relational schema.

4 **Physical Design**

- Model user-level data organization in a specific DBMS (and data model) Account for deployment environment and performance requirements.
- **Key Questions**: Which indexes? What storage engine? How to partition?
- **Deliverable:** Annotated schema + storage spec.

3 Conceptual Data Modeling: Translating Ideas into Structure

Before drawing an Entity-Relationship Diagram (ERD), you must first define the **Conceptual Data Model** itself. While an ERD is the visual chart or diagram, conceptual data modeling is the high-level design process that identifies *what* core concepts exist in a domain and *how* they connect.

📖 DEFINITION 2: **Conceptual Data Model**

is the general language, rules, and notation framework used to model data conceptually (such as the **Entity-Relationship Model (ERM)** or **UML**).

📖 DEFINITION 3: **Conceptual Schema**

A description of the *structure* of a database at a level that is independent of any specific DBMS, storage format, or programming language. It answers: *"What data exists, and how is it connected?"*— not *"How is it stored?"*

3.1 The Purpose of Conceptual Data Modeling

1. **High-Level Abstraction:** It captures essential business concepts without being tied to specific technology, software engines, or database types (SQL vs. NoSQL).
2. **Stakeholder Bridge:** It serves as a shared vocabulary between technical developers and non-technical business managers to ensure everyone agrees on basic terms.
3. **Scope Definition:** It sets clear boundaries on what information the system will track and what it will ignore.

3.2 Core Building Blocks

Every conceptual model relies on three fundamental elements:

- **Entities (Objects of Interest):** Real-world subjects, places, or events you need to store information about.
  - *Example:* `Customer`, `Order`, `Product`, `Employee`.

- **Attributes (Properties & Characteristics):** Essential traits that describe each entity.
  - *Example:* A `Customer` entity has attributes like `Name` and `EmailAddress`.
- **Relationships (Business Rules & Connections):** Statements describing how entities interact with one another.
  - *Example:* A `Customer` **places** an `Order`.

3.3 The Conceptual Design Process

To move from raw business requirements to a formal model, follow these initial steps:

- **Extract Nouns & Verbs from Requirements:** Read through your business rules. Nouns usually map to **Entities** or **Attributes**, while verbs map to **Relationships**.
  - *Requirement Text: "A student enrolls in multiple courses."*
  - *Entities:* `Student`, `Course`
  - *Relationship:* **Enrolls In**
- **Determine Cardinality & Business Constraints:** Decide how many instances of one entity can relate to another (e.g., 1-to-1, 1-to-Many, Many-to-Many).
- **Draft the Visual Representation:** Once these rules and entities are clearly defined, you translate them into visual notation using an **ER Diagram (ERD)**.

1 **Extract Nouns & Verbs from Requirements**

Read through your business rules. Nouns usually map to **Entities** or **Attributes**, while verbs map to **Relationships**.

- *Requirement Text: "A student enrolls in multiple courses."*
- *Entities:* `Student`, `Course`
- *Relationship:* **Enrolls In**

2 **Determine Cardinality & Business Constraints**

Decide how many instances of one entity can relate to another (e.g., 1-to-1, 1-to-Many, Many-to-Many).

3 **Draft the Visual Representation**

Once these rules and entities are clearly defined, you translate them into visual notation using an **ER Diagram (ERD).**

4 What Is an ER Diagram?

ER modeling was introduced by **Peter Chen in 1976**. Its core contribution was giving database designers a single, shared language to communicate with both business stakeholders (who understand the problem domain) and software engineers (who will implement the solution).

> 📖 DEFINITION 4: **Entity-Relationship (ER) Diagram**
>
> A graphical representation of a conceptual schema—a high-level, implementation-independent description of the data a system must store.

> 📖 DEFINITION 5: **Conceptual Schema**
>
> A description of the *structure* of a database at a level that is independent of any specific DBMS, storage format, or programming language. It answers: *"What data exists, and how is it connected?"*— not *"How is it stored?"*

4.1 ER Notations Overview

An ER diagram is a graphical language, and like any language it comes in different dialects. Three notations are widely used; you will encounter all three in textbooks and professional tools.

Table 6: *Comparison of standard Entity-Relationship (ER) diagram notations, including creator origins and primary use cases.*

Notation	Introduced by	Where you see it
Chen	Peter Chen, 1976	University textbooks, academic exams, this course
Min-Max (Structural Constraints)	Various authors	Formal specifications needing exact minimum and maximum counts
Crow's Foot	Industry convention	Design tools — MySQL Workbench, Lucidchart, draw.io, Visio, ERwin

All three notations express the same concepts — entities, attributes, relationships, cardinality, and participation — using different visual symbols. You do not need to master all three right now. The important thing is to know they exist so you can recognise a diagram regardless of the notation it uses.

4.2 Chen Notation

4.2.1 Chen Notation — Symbol Reference

Chen notation is the academic standard you will use for all diagrams in this course. The table below is your reference for every symbol.

Table 7: *Chen Notation — Shape Reference*

Concept	Shape	Description
Strong Entity	Single rectangle	An object with independent existence and its own key
Weak Entity	Double rectangle	An object that depends on another entity for identification
Relationship	Single diamond	An association between two or more entities
Identifying Relationship	Double diamond	The relationship that identifies a weak entity
Attribute	Ellipse (oval)	A property of an entity or relationship
Key Attribute	Ellipse with underlined name	Uniquely identifies each entity instance
Multivalued Attribute	Double ellipse	Can hold multiple values per instance
Derived Attribute	Dashed ellipse	Computed from another stored attribute
Composite Attribute	Ellipse → child ellipses	Built from smaller sub-attributes
Total Participation	Double line	Every instance of the entity must participate
Partial Participation	Single line	Some instances may not participate
Cardinality label	1, N, or M on the line	Placed near each entity; shows the maximum

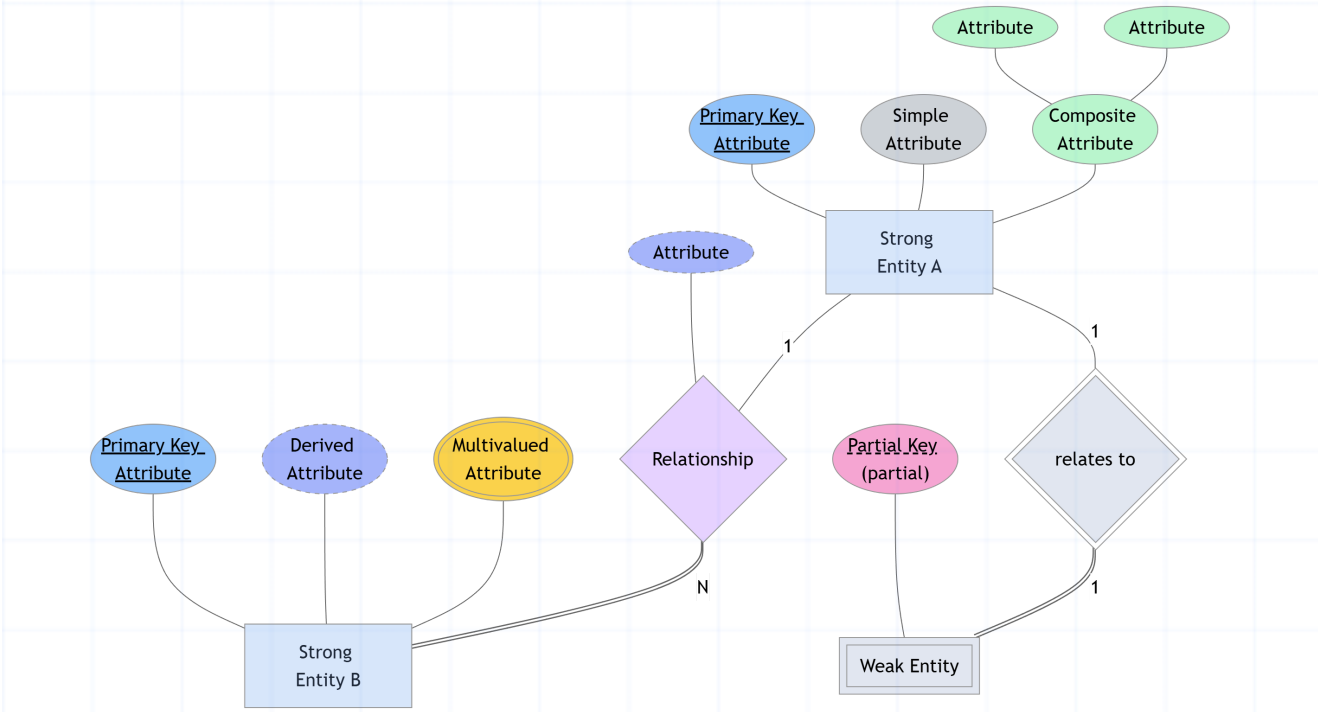


Diagram 1: Chen Notation — Shape Reference

4.2.2 Entities

4.2.2.1 Strong Entity

A strong entity type (or simply entity type) is a real-world object or concept that:

1. Has an independent existence — it does not depend on another object to exist in the database.
2. Has a set of attributes that describe it.
3. Has at least one attribute (or combination) that uniquely identifies each instance — its key attribute.

Notation: A **single rectangle** labeled with the entity name (all caps by convention).

📝 EXAMPLE 2: **STUDENT entity with its attributes in Chen notation**

`Student` is the entity (rectangle), `ID` is the key attribute (underlined), and `NAME`, `GENDER`, `DOB` are simple attributes (ovals connected by lines).

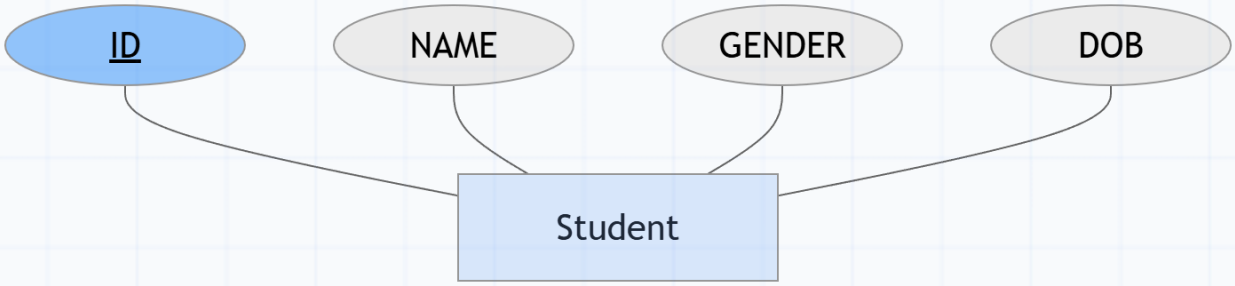


Diagram 2: Entity STUDENT with attributes ID (key), NAME, GENDER, and DOB

4.2.2.2 Weak Entity

A **weak entity type** has **no key attribute of its own**— it cannot be uniquely identified without reference to another entity (its **owner**).

📖 DEFINITION 6: **Partial Key**

The attribute(s) of a weak entity that, combined with the owner entity's key, uniquely identify a weak entity instance.

Table 8: *Comparison of Key Features and Notation Differences Between Strong and Weak Entities*

Feature	Strong Entity	Weak Entity
Has own key?	Yes	No — only a partial key
Identified by	Own primary key	Owner's key + partial key
Rectangle notation	Single	Double
Identifying relationship	Single diamond	Double diamond
Participation in identifying rel.	May vary	Always total

Key characteristics include:

1. **Identification**: Identified by combining the owner entity's primary key with its own partial key (discriminator).
2. **Notation**: Drawn as a double rectangle in Chen notation.
3. **Relationship**: Connected to its owner through an identifying relationship (drawn as a double diamond).
4. **Participation**: Always has total participation in its identifying relationship, meaning a weak entity instance cannot exist in the database without an owner.

EXAMPLE 3:

Room is a weak entity identified by Hotel (owner) + roomNumber (partial key, dashed underline). The HAS relationship is the identifying relationship (double diamond in full Chen notation).

- `Hotel` is the **strong entity**— it has its own key (`hotelID` , underlined).
- `Room` is the **weak entity**— `roomNumber` alone is not unique across all hotels (two hotels can both have room 101). It is identified by the combination `(hotelID, roomNumber)` .
- `HAS` is the **identifying relationship**(1:N — one hotel has many rooms).

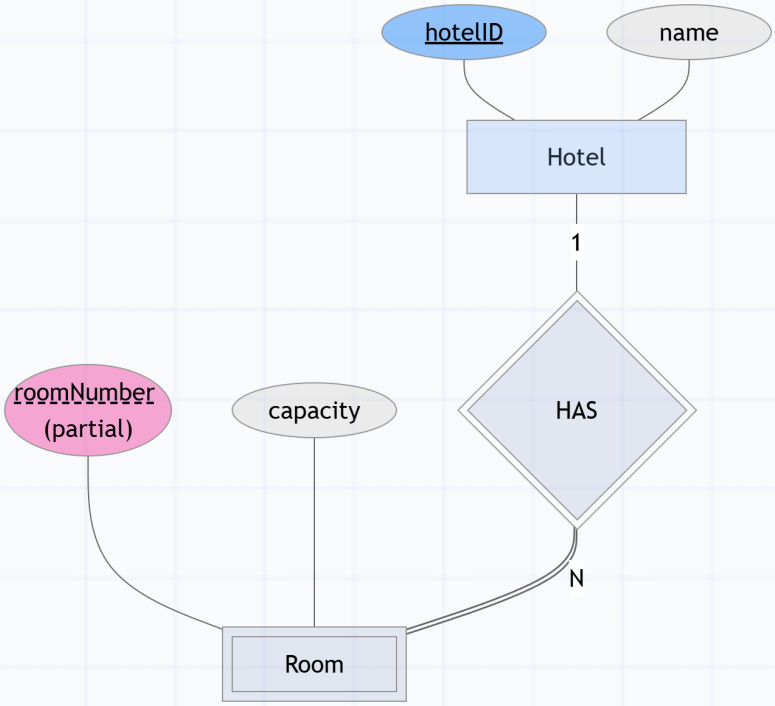


Diagram 3: Hotel (strong) and Room (weak entity) in Chen notation

EXAMPLE 4: Local Order Numbers (Order is a WEAK Entity)

- **The Setup:** Each location has its own Point of Sale (POS) system. Location A generates Order 1, 2, 3... Location B *also* generates Order 1, 2, 3...
- **The Problem:** If I say "Look at Order #1", you would ask, "*Which location's Order #1?*"
- **The Database Reality:** The `Order_Number` alone is not unique. To uniquely identify the record, the database **must** use a composite key: `(Location_ID, Local_Order_Number)` .

- **The Conclusion:** Because the `Order` relies on the `Location`'s ID to form its primary key, it is **Identification Dependent**. Therefore, **Order is a Weak Entity**. (The `Order_Number` acts as the *partial key* or *discriminator*).

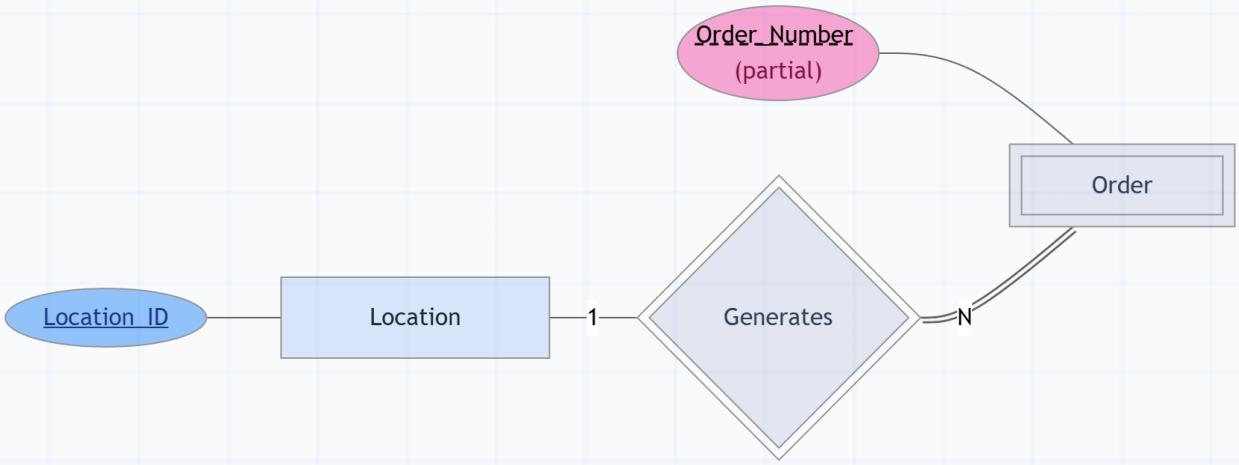


Diagram 4: Local Order Numbers (Order is a WEAK Entity)

📝 EXAMPLE 5: **Centralized Order Numbers (Order is a STRONG Entity)**

- **The Setup:** All locations connect to one central server to generate order IDs. Location A gets Order #1001, Location B gets Order #1002. The numbers never repeat across locations.
- **The Reality:** If I say "Look at Order #1001", you can find it immediately. You don't need to know the location to find the order, because `Order #1001` only exists once in the entire system.
- **The Database Reality:** The Primary Key is simply `Order_ID`. It is globally unique.
- **The Conclusion:** Because the `Order` has its own independent, globally unique identifier, it is **Identification Independent**. Therefore, **Order is a Strong Entity**.
- *(Note: The order still **must** belong to a location, so it has Total Participation / Existence Dependency, but as we learned with the passport, that does NOT make it weak!)*

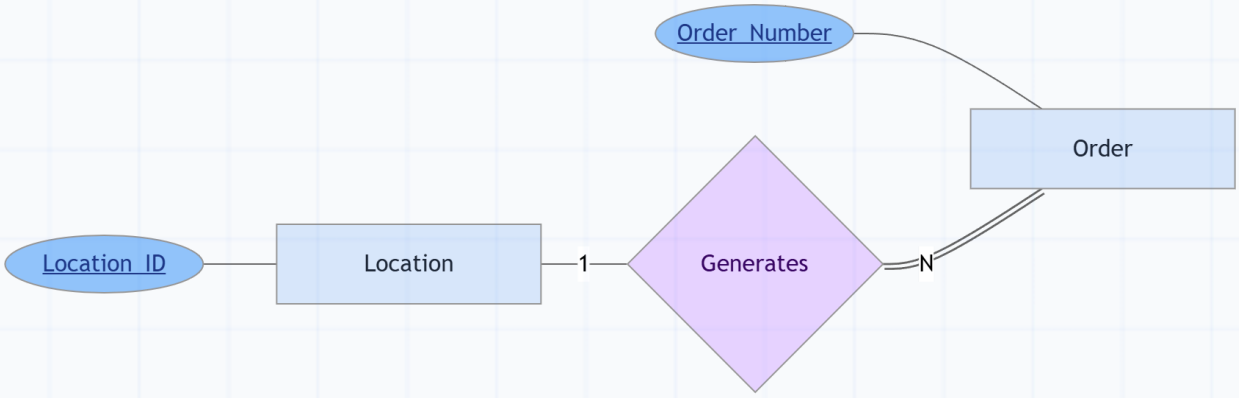


Diagram 5: Centralized Order Numbers (Order is a STRONG Entity)

4.2.3 Attributes

📖 DEFINITION 7: **Attribute**

A property that describes an entity type or relationship type.

Table 9: *Summary of ER Diagram Attribute Types, Descriptions, Chen Notation Visual Representations, and Examples*

Type	Description	Chen Shape	Example
Simple	Atomic, cannot be divided further	Plain oval	Salary, Gender
Composite	Built from smaller sub-attributes	Oval feeding child ovals	Name → (Fname, Minit, Lname)
Multivalued	Can hold multiple values per entity instance	Double oval	Phone_Numbers, Dept_Locations
Derived	Computed from another stored attribute; not physically stored	Dashed oval	Age derived from Bdate
Stored	Permanently stored; may be a source for derived attributes	Plain oval	Bdate
Key	Uniquely identifies each entity instance	Oval with underlined text	Ssn, Dnumber
Partial key	Identifies weak entity instances within the owner; not globally unique	Oval with dashed underline	Name (of DEPENDENT)

4.2.3.1 Simple (Atomic) Attribute

📖 DEFINITION 8: **Simple (Atomic) Attribute**

An attribute that cannot be divided into smaller meaningful parts.

📝 EXAMPLE 6: **Simple attributes — Product entity: every oval holds one indivisible value**

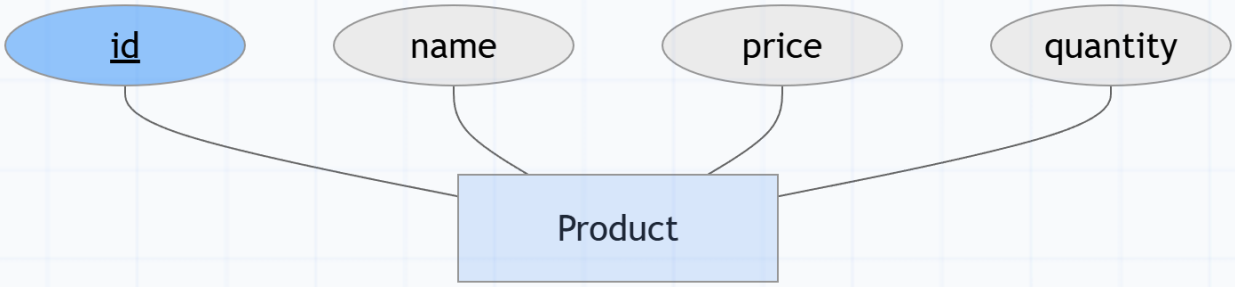


Diagram 6: Simple attributes — Product entity: every oval holds one indivisible value

4.2.3.2 Composite Attribute

📖 DEFINITION 9: **Composite Attribute**

An attribute made up of smaller, meaningful sub-attributes.

📝 EXAMPLE 7: **Composite attributes — Employee FullName and Address each decompose into sub-attributes**

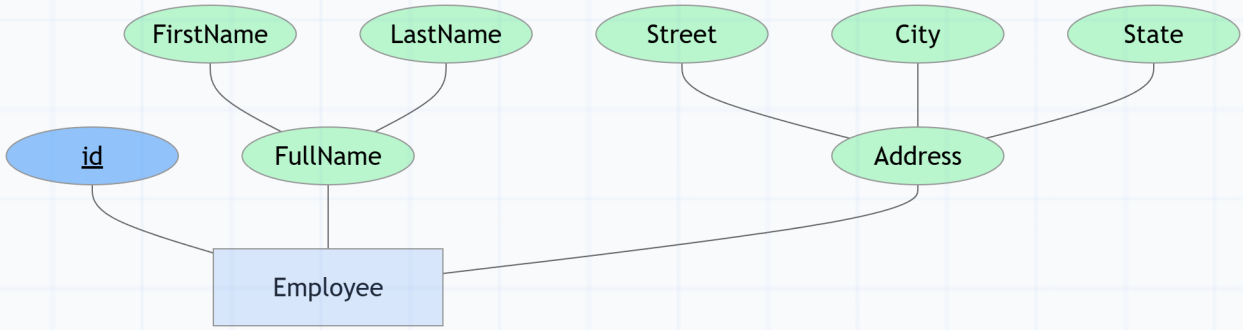


Diagram 7: Composite attributes — Employee FullName and Address each decompose into sub-attributes

4.2.3.3 Single-Valued Attribute

📖 DEFINITION 10: **Single-Valued Attribute**

An attribute that holds exactly one value for each entity instance.

📝 EXAMPLE 8: **Single-valued attributes — each User has exactly one id, one name and one email**

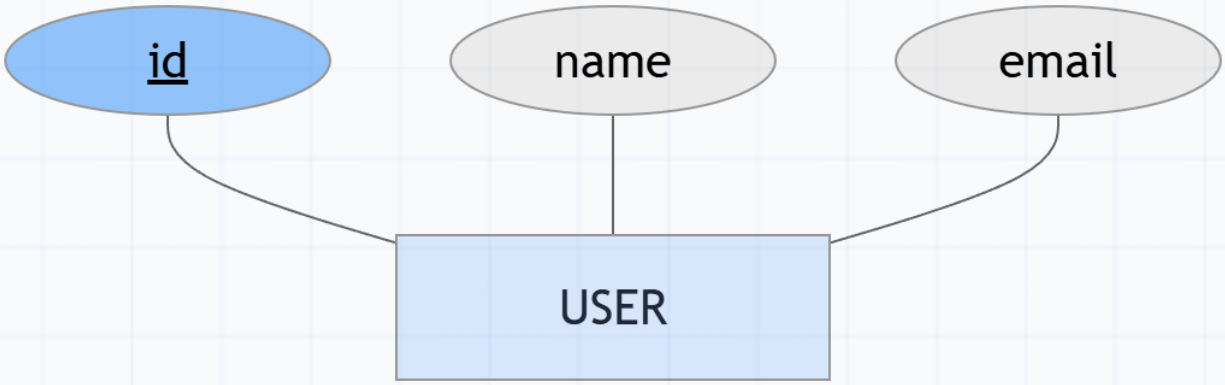


Diagram 8: Single-valued attributes — each User has exactly one id, one name and one email

4.2.3.4 Multi-Valued Attribute

📖 DEFINITION 11: **Multi-Valued Attribute**

An attribute that can hold more than one value for the same entity instance.

EXAMPLE 9: Multi-valued attributes — Addresses and Skills drawn as double ovals because one employee may have several

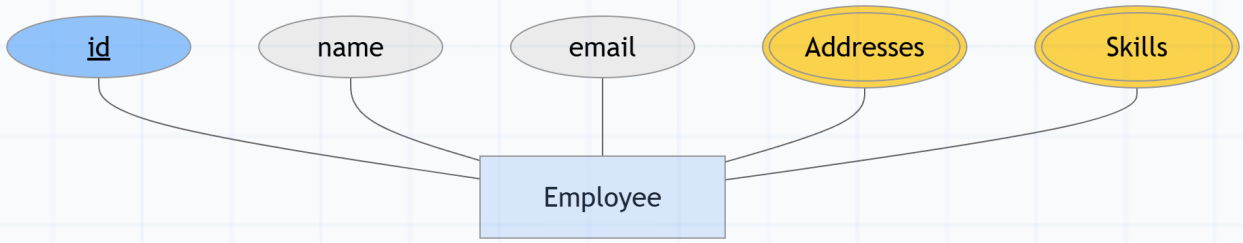


Diagram 9: Multi-valued attributes — Addresses and Skills drawn as double ovals because one employee may have several

4.2.3.5 Derived Attribute

DEFINITION 12: Derived Attribute

An attribute computed from another stored attribute rather than physically stored in the database.

EXAMPLE 10: Derived attribute — Age (dashed oval) is computed from the stored DOB attribute

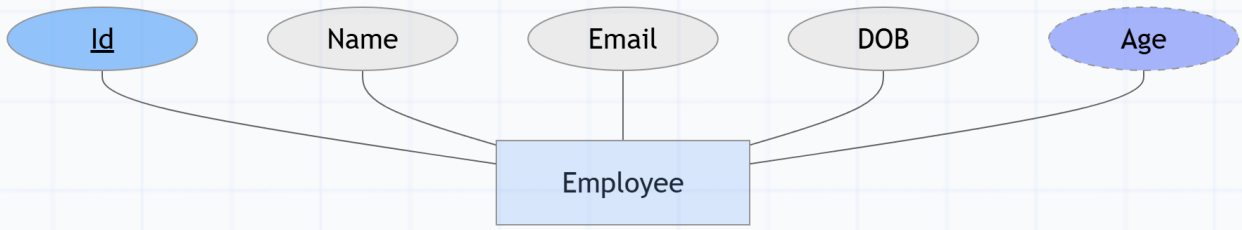


Diagram 10: Derived attribute — Age (dashed oval) is computed from the stored DOB attribute

In the example, `Age` (dashed oval) is derived from the stored `DOB` attribute.

4.2.3.6 Key Attribute

DEFINITION 13: Key Attribute

An attribute that uniquely identifies each entity instance.

The ER Key Rule

Every strong entity type must have exactly one key attribute(or one composite key attribute group) that uniquely identifies each instance. You cannot draw a valid ER diagram for a strong entity without specifying its key.

Weak entities do not have their own key — instead they have a **partial key** that, combined with the owner entity's key, forms a composite identifier.

What about Foreign Keys?

Foreign keys do **not** appear in ER diagrams. They are a relational concept — a column in one table that references the primary key of another table. Foreign keys are introduced in Chapter 5 and created automatically by the ER-to-relational mapping rules.

4.2.3.6.1 Key Attribute (Strong Entity)

A **key attribute** is drawn as an oval with its name **underlined**. It guarantees that no two instances of the entity have the same value for that attribute.

- One entity can have **only one key attribute**(or one composite key group) at the ER level.
- If two candidate identifiers exist (e.g., both `SSN` and `Email` are unique), you underline **one** in the ER diagram — the one you choose as the primary identifier. The other becomes an alternate key at the relational level.
- The key attribute in the ER diagram maps directly to the **primary key** column(s) when the entity becomes a relational table.

4.2.3.6.2 Partial Key (Weak Entity)

A **partial key** is drawn with a **dashed underline**. It is *not* globally unique — it is only unique within the set of instances owned by the same owner entity.

📖 DEFINITION 14: **Partial Key**

The attribute(s) of a weak entity that, combined with the owner entity's key, uniquely identify a weak entity instance.

Note

- The full identifier of a weak entity instance is: **owner's primary key + partial key**.
- This combination maps to a **composite primary key** in the relational table.

4.2.4 Relationships

📖 DEFINITION 15: **Relationship**

An association or connection among two or more entity types. In Chen notation, it is represented visually by a **single diamond** labeled with a verb or verb phrase describing the association.

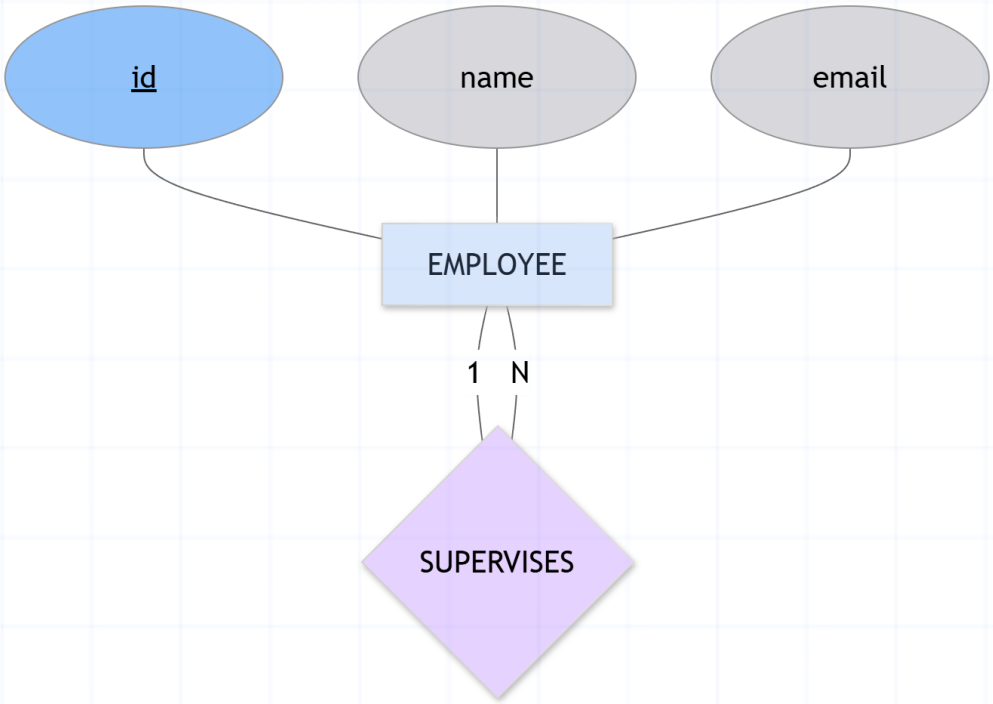
4.2.4.1 Degree of a Relationship

📖 DEFINITION 16: **Degree of a Relationship**

> The number of participating entity types in a relationship (e.g., Unary/1, Binary/2, Ternary/3).

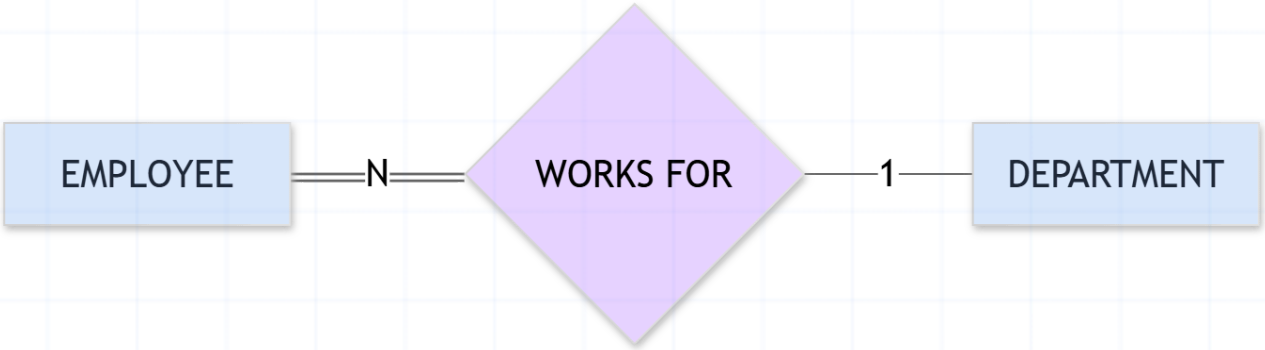
The three degrees are:

- **Unary (Recursive):** A single entity type associated with itself (e.g., an `EMPLOYEE` supervising another `EMPLOYEE`).



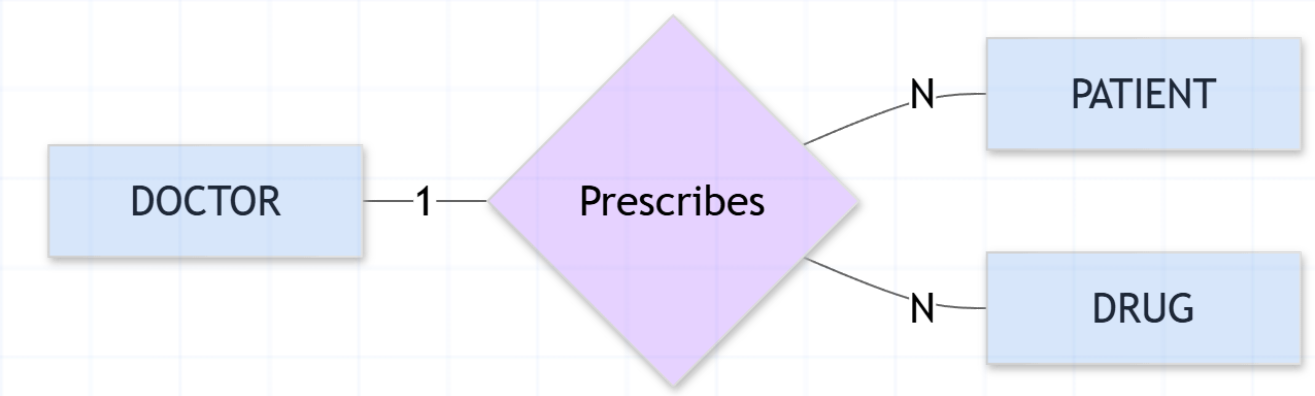
A unary (recursive) relationship where a single entity type (EMPLOYEE) interacts with itself. One employee acts as a supervisor for N supervisee employees.

- **Binary:** Connects two entity types (e.g., `EMPLOYEE` works for `DEPARTMENT`).



A binary relationship connecting two distinct entity types (EMPLOYEE and DEPARTMENT), where many employees work for one department.

- **Ternary:** Involves three entity types simultaneously.



A 1:N:N ternary relationship (Prescribes) connecting three entity types: DOCTOR, DRUG, and PATIENT.

> **Note**
>
> Most real-world relationships are **binary**. Before using a ternary relationship, ask whether it can be decomposed into two binary ones without losing information.

4.2.4.2 Cardinality Ratios

> 📖 DEFINITION 17: **Cardinality Ratio**
>
> Specifies how many instances of one entity can be associated with one instance of another entity (1:1, 1:N, M:N).

4.2.4.2.1 One-to-One (1:1)

Each instance of entity A is associated with **at most one** instance of entity B, and vice versa.

> 📝 EXAMPLE 11: **1:1 relationship — each Person has at most one Passport; each Passport belongs to exactly one Person**
>
> **Requirements:**
>
> - Each person has **at most one** passport — not everyone has a passport.
> - Each passport belongs to **exactly one** person — every passport has an owner.

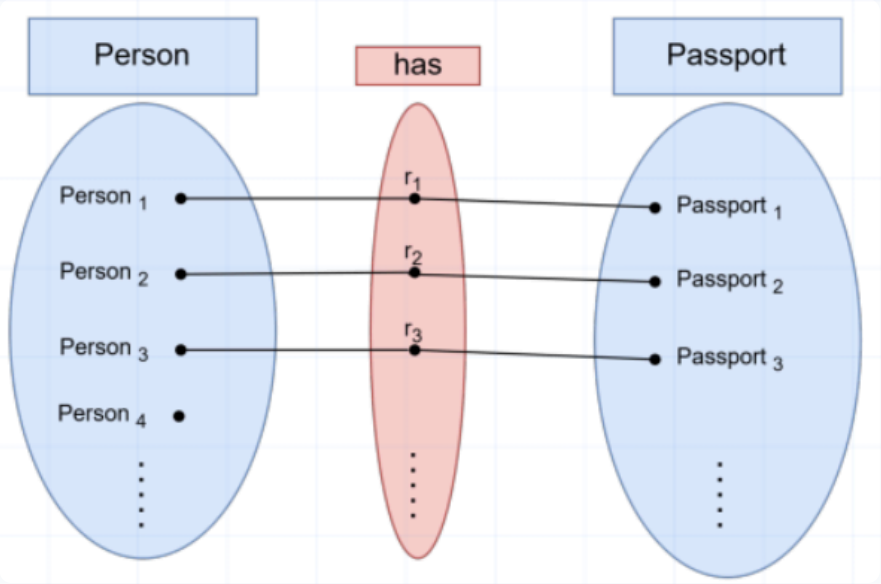


Figure 8: 1:1 relationship — each Person has at most one Passport; each Passport belongs to exactly one Person

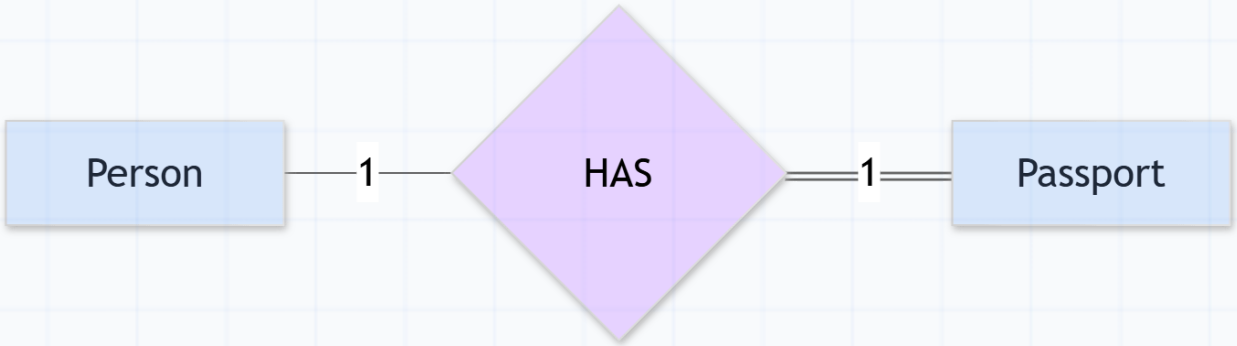


Diagram 11: 1:1 relationship — each Person has at most one Passport; each Passport belongs to exactly one Person

Reading: "A person may have zero or one passport; a passport must be owned by exactly one person."

Why Passport is modeled as a strong entity rather than a weak entity

The reason a **Passport** is modeled as a **strong entity** rather than a weak entity is precisely because it has its own unique identifier (the **Passport Number**) that can uniquely identify each passport record on its own, without needing to borrow an identifier from the `Person` entity.

Strong Entity vs. Weak Entity (It's about the Primary Key)

- **Strong Entity**: Has its own **primary key** (a unique identifier). It can be uniquely identified in the database independently.
- **Weak Entity**: **Does not have its own primary key**. It can only be uniquely identified by combining its partial identifier (discriminator) with the primary key of another entity (the owner entity).
- *Application to Passport*: A `Passport Number` is globally unique (or unique per country). Therefore, `Passport` has its own primary key and is a **strong entity**.

4.2.4.2.2 One-to-Many (1:N)

One instance of entity A is associated with **many** instances of entity B, but each B instance is associated with at most one A.

EXAMPLE 12: 1:N relationship — one Department employs many Employees; each Employee belongs to exactly one Department

Requirements:

- A department can exist **without** employees (e.g., a newly created department with no staff yet) →**partial** participation of Department in the relationship.
- An employee **must** be assigned to a department →**total** participation of Employee in the relationship.

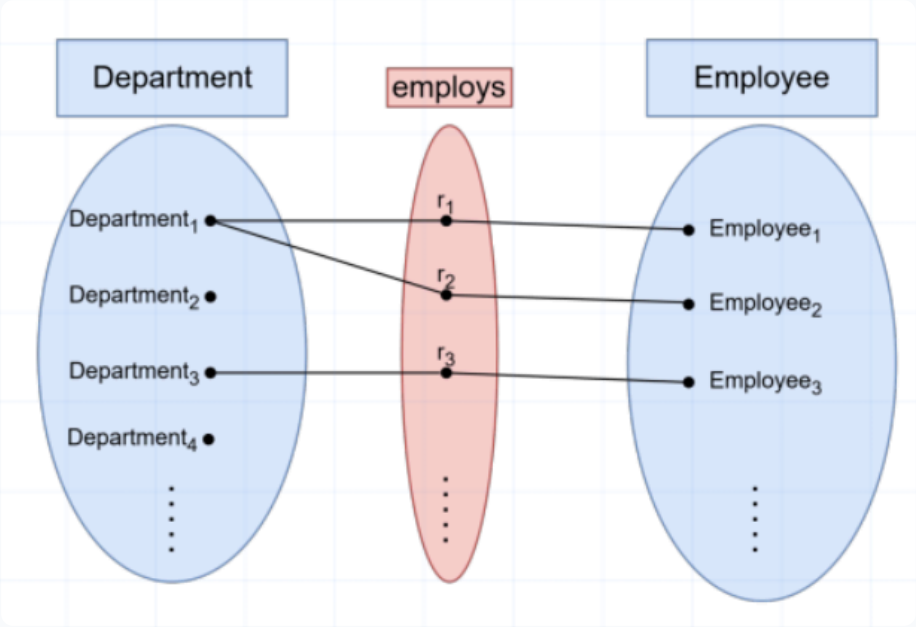


Figure 9: 1:N relationship — one Department employs many Employees; each Employee belongs to exactly one Department

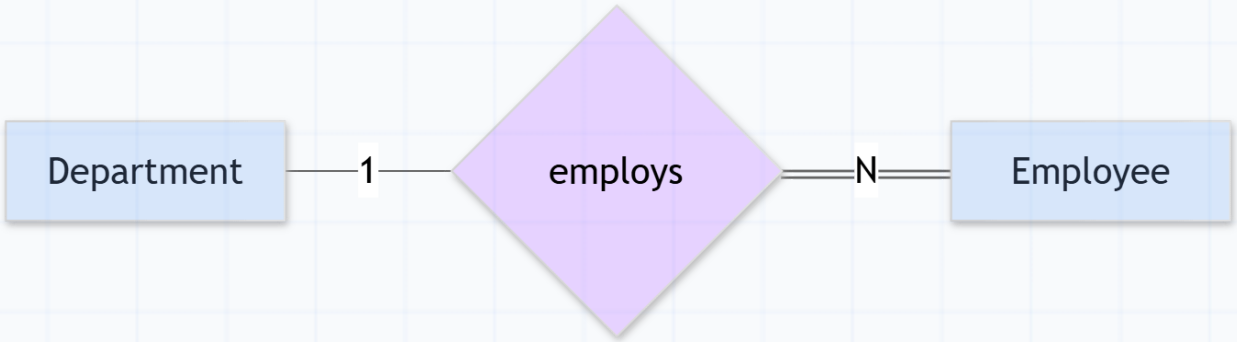


Diagram 12: 1:N relationship — one Department employs many Employees; each Employee belongs to exactly one Department

Reading:

- **Left → Right (Department → Employee):**One department employs zero or more employees.
- **Right → Left (Employee → Department):**Each employee works for exactly one department.

4.2.4.2.3 Many-to-Many (M:N)

Each instance of entity A can be associated with **many** instances of entity B, and each B can be associated with many A's.

EXAMPLE 13: M:N relationship — many Students enroll in many Courses; many Courses have many Students

Requirements:

- One student can enroll in multiple courses, and one course can have multiple students enrolled.
- A student can exist in the system without currently being enrolled in any course (e.g., a newly admitted student).
- A course can exist in the system without any enrolled students (e.g., a newly created course section).

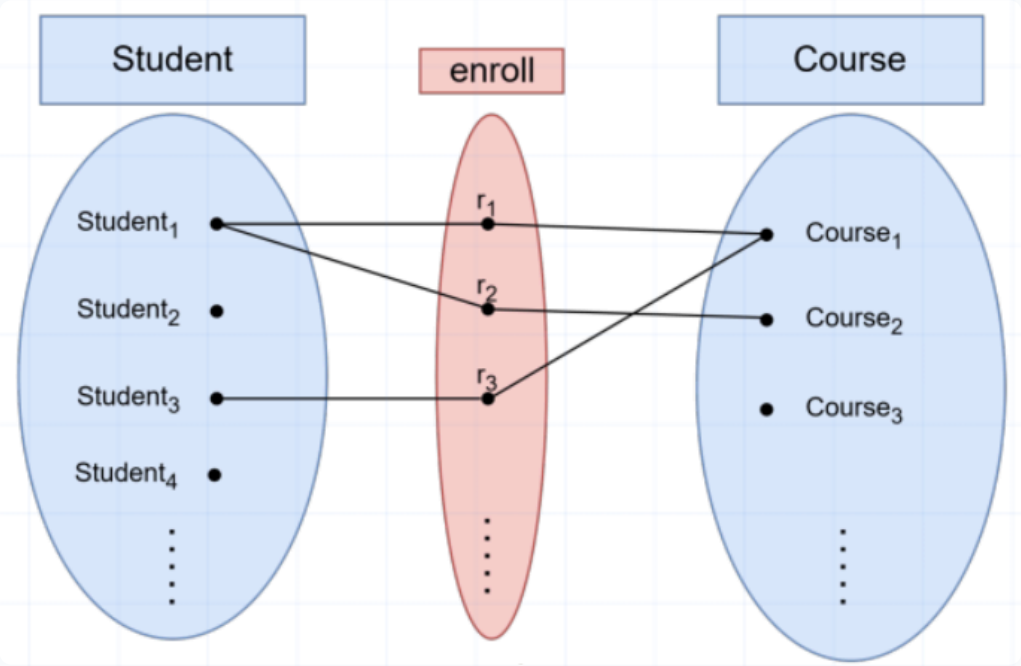


Figure 10: M:N relationship — many Students enroll in many Courses; many Courses have many Students

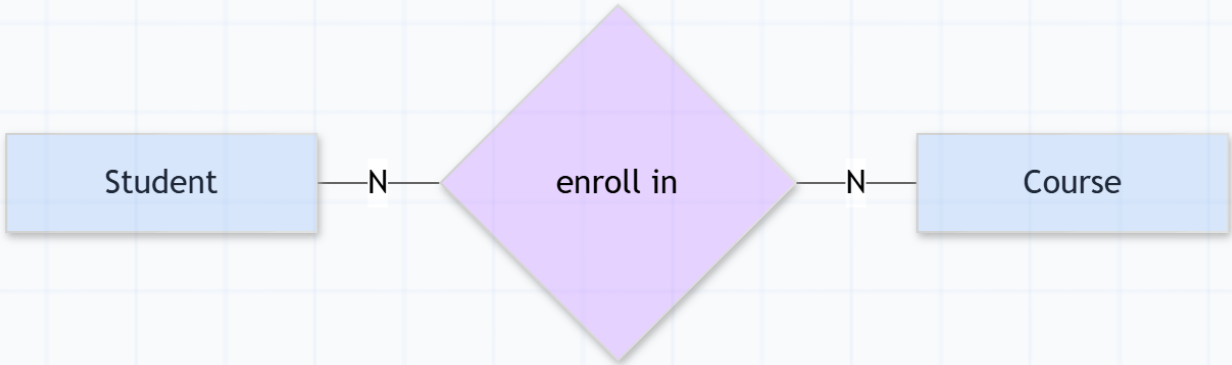


Diagram 13: M:N relationship — many Students enroll in many Courses; many Courses have many Students

Reading:

- **Left → Right (Student → Course):**A student may enroll in zero or more courses.
  Right → Left (Course → Student):A course may have zero or more students.

4.2.4.3 Participation Constraints

📖 **DEFINITION 18: Participation Constraint**

Specifies whether every instance of an entity must participate in a relationship (Total) or if participation is optional (Partial).

Table 10: *Participation Constraints Visual and Conceptual Summary*

Line type	Name	Meaning
Single line —	Partial participation	Some instances may not participate
Double line ═	Total participation	Every instance must participate

📝 **EXAMPLE 14: Department managed By Employee**

- **Department** side →**double line (total)**: every department *must* be managed by exactly one employee.
- **Employee** side →**single line (partial)**: not every employee is a manager — an employee *may* manage a department, but doesn't have to.

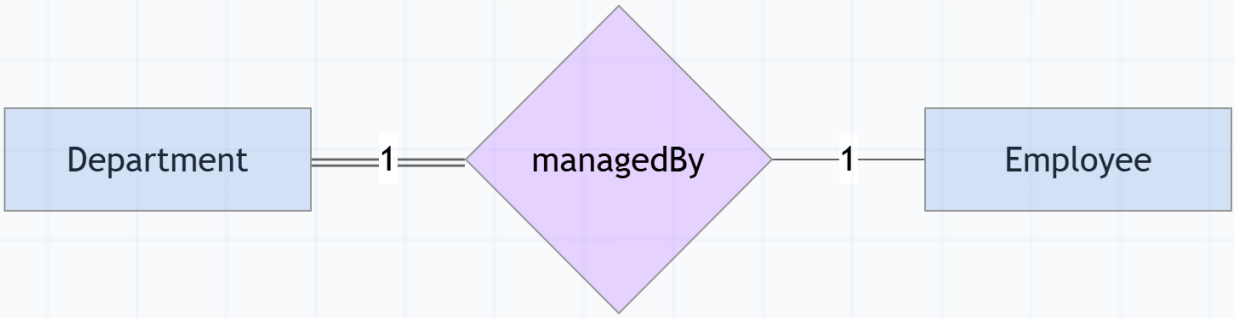


Diagram 14: Department managed By Employee

ⓘ **Note**

Total participation means the entity **cannot exist** in the database without being connected to this relationship. A department record cannot be inserted unless it has a managing employee assigned to it.

4.2.4.4 Relationship Attributes

📖 **DEFINITION 19: Relationship Attribute**

Data or a value that belongs specifically to the pairing of entities rather than to either entity alone.

📝 **EXAMPLE 15: WORKS_ON relationship attribute**

`Hours` in *WORKS_ON* is a **relationship attribute**: it is the number of hours *this employee* spends on *this project*.

- Storing it with `EMPLOYEE` is wrong: hours on which project?
- Storing it with `PROJECT` is wrong: hours by which employee?
- It belongs to the *combination* `(Employee, Project)` — i.e., the relationship.

In Chen notation, relationship attributes are drawn as ovals attached to the diamond.

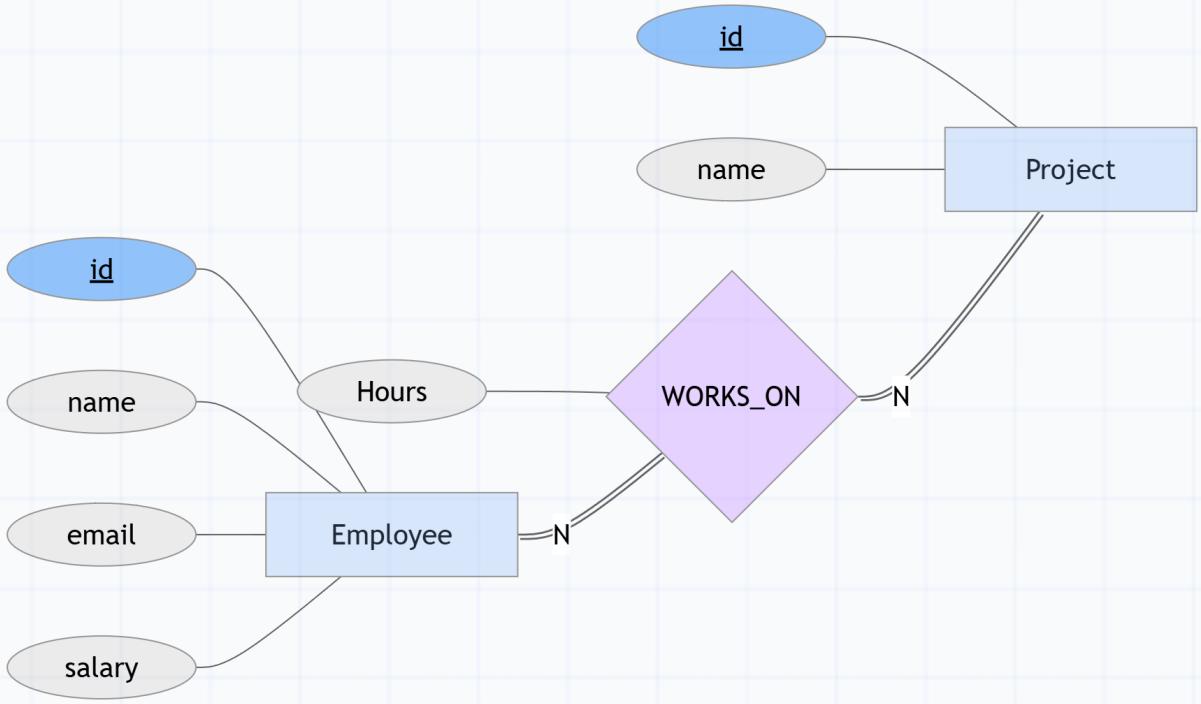


Diagram 15: WORKS_ON relationship attribute — Hours is attached to the WORKS_ON diamond because it belongs to the (Employee, Project) pairing, not to either entity alone

4.2.4.5 Recursive (Unary) Relationships

A relationship where the **same entity type** participates more than once, in different **roles**.

EXAMPLE 16: **Recursive relationship : Supervisor is a role of Employee**

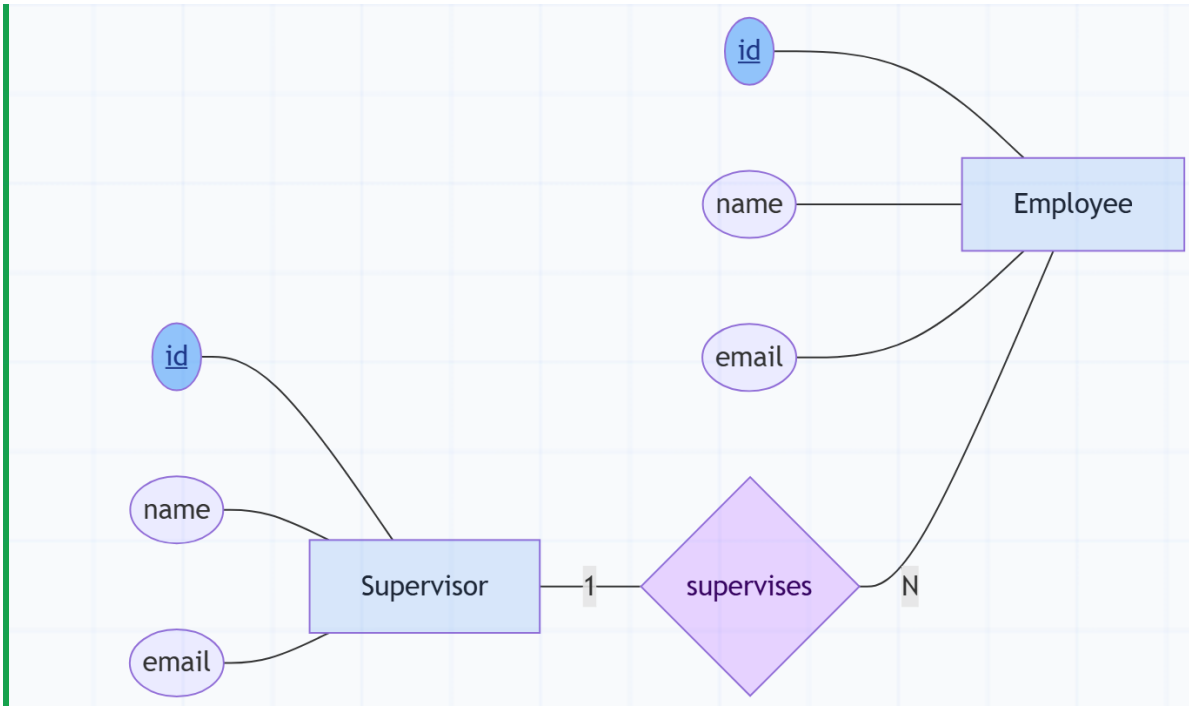


Diagram 16: Supervisor is a role of Employee (wrong)

Why the first diagram is wrong: `Supervisor` is not a different kind of thing from `Employee` — a supervisor *is* an employee who happens to manage others. Drawing two separate entity boxes implies two different tables in the database, which is incorrect.

> **Note**
>
> **Partial Participation on Both Ends:** Use single lines for both connections because an employee can exist without supervising anyone (regular workers), and a top-level manager can exist without having a supervisor above them.

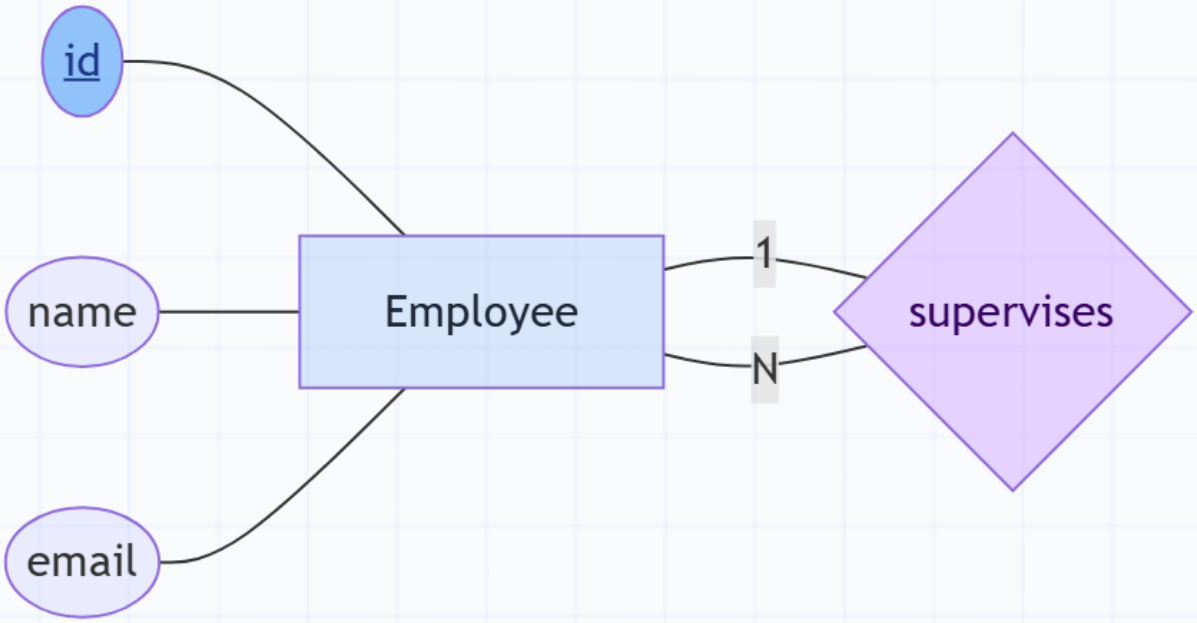


Diagram 17: Supervisor is a role of Employee (correct)

> **Why the second diagram is correct:** A single `Employee` entity with a **recursive** `SUPERVISION` relationship back to itself, with two labelled roles: `Supervisor` (1 side) and `Supervisee` (N side). One entity maps to one table — the role is just a label on the line.

4.3 Worked Example: The Company Database

> **Note**
>
> This section builds the **Company Database ER diagram** incrementally — one requirement at a time. For each step, the relevant sentence from the requirements is highlighted, followed by the updated diagram and an explicit note on any assumption made.

4.3.1 Requirements

The company has several **departments**. Each department has a unique **name** and a **number**. A department is managed by exactly one **employee**, who is called the **manager**. The manager has a **start date** for their managerial role. A department may have several **locations**.

Each employee has a unique **social security number (SSN)**, a **first name**, **middle initial**, **last name**, **date of birth**, **address**, **gender**, and **salary**. An employee's **age** can be derived from their date of birth and does not need to be stored separately. Each employee works in exactly one department but may work on several **projects**. We track the **number of hours** per week each employee works on each project. Each employee may be supervised by at most one other employee.

Each project has a unique **name**, a **number**, and a single **location**. Each project is controlled by exactly one department.

Employees may have **dependents**. Each dependent has a **name**, **gender**, **date of birth**, and **relationship** to the employee.

4.3.2 Step 1 — EMPLOYEE Entity

> **From Requirements**
>
> Each employee has a unique social security number (SSN), a first name, middle initial, last name, date of birth, address, gender, and salary. An employee's age can be derived from their date of birth and does not need to be stored separately.

We start with the most described entity: EMPLOYEE.

- `SSN` →**key attribute**(underlined oval) — uniquely identifies each employee
- `Name` →**composite attribute**— decomposes into `Fname` , `Minit` , `Lname`
- `Bdate` → stored attribute — the value is physically kept in the database

- `Age` →**derived attribute**(dashed oval) — computed from `Bdate`; never stored
- `Address`, `Gender`, `Salary` → simple, single-valued attributes

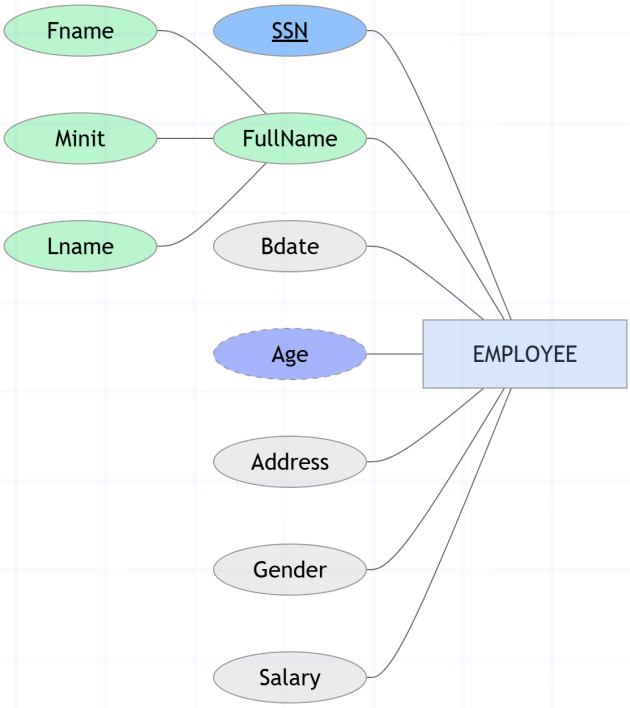


Diagram 18: Step 1 — EMPLOYEE entity with all its attributes

4.3.3 Step 2 — DEPARTMENT Entity + MANAGES Relationship

> **From Requirements**
>
> The company has several departments. Each department has a unique name and a number. A department may have several locations. A department is managed by exactly one employee, who is called the manager. The manager has a start date for their managerial role.

We start with the most described entity: EMPLOYEE.

- `DEPARTMENT` is a strong entity; `Number` is its key attribute
- `Locations` →**multivalued attribute**(double oval) —*"may have several"*
- `Start_date` →**relationship attribute**on MANAGES — it belongs to the (Employee, Department) pairing, not to either entity alone
- **Department side of MANAGES:***"exactly one"*→**total**participation, max = 1 (double line, label 1)
- **Employee side of MANAGES:**The requirement does not say every employee is a manager. "Manager" is a*role*played by some employees, not a separate entity type.

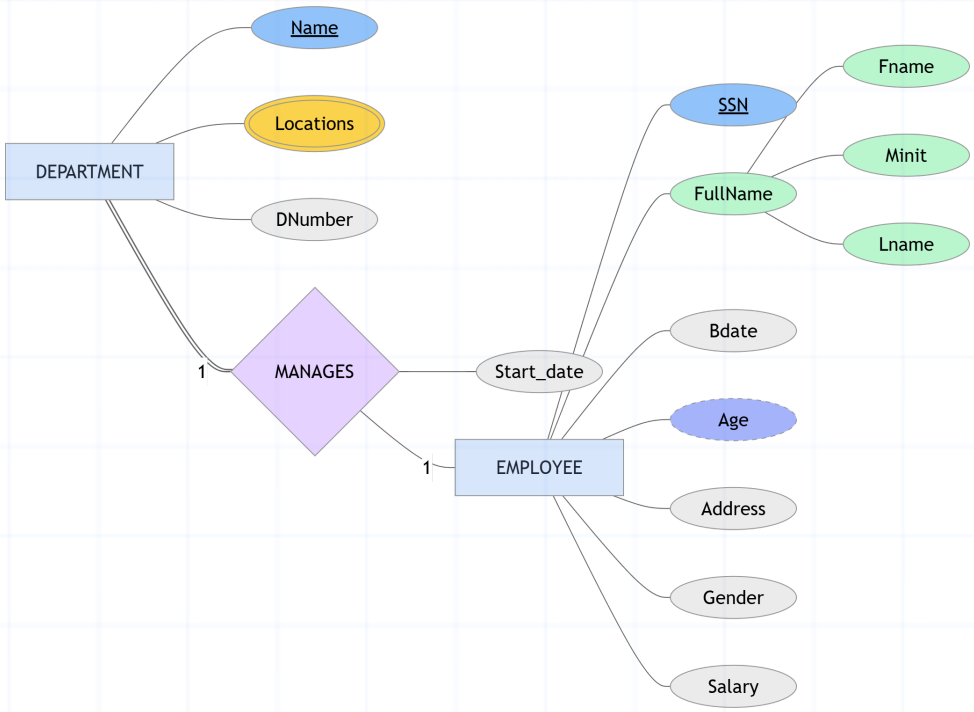


Diagram 19: Step 2 — DEPARTMENT Entity + MANAGES Relationship

4.3.4 Step 3 — WORKS_FOR Relationship

> **From Requirements**
>
> Each employee works in exactly one department.

WORKS_FOR connects EMPLOYEE and DEPARTMENT with cardinality N:1.

- **Employee side:** *"exactly one"* →**total** participation (double line), max = 1
- **Department side:** The requirement only states where employees go; it says nothing about whether a department must have employees.

> **Assumption — WORKS_FOR (Department side)**
>
> **Stated:** Every employee must belong to exactly one department → Employee side is **total**.
>
> **Assumption:** The diagram shows a **single line** on the Department side, meaning a department can exist with zero employees assigned to it → Department side is **partial**, min = 0, max = N.
>
> *This is a design choice. You could require every department to have at least one employee by making their side total (double line).*

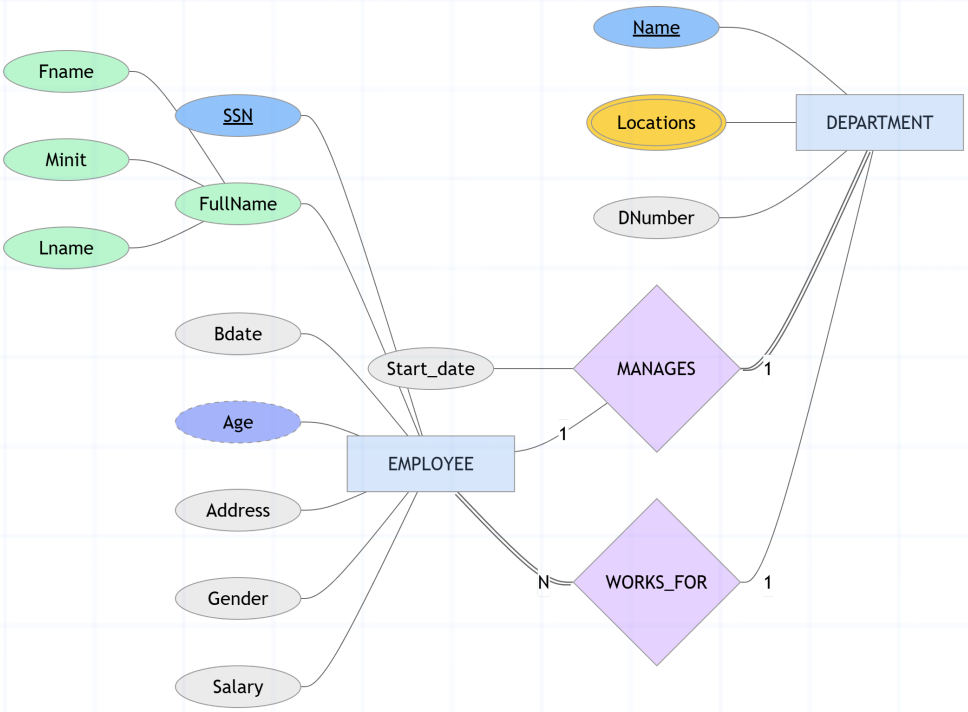


Diagram 20: Step 3 — WORKS_FOR Relationship

4.3.5 Step 4 — PROJECT Entity

> **From Requirements**
>
> Each project has a unique name, a number, and a single location.

PROJECT is a strong entity with three attributes.

- `Number` →**key attribute**(underlined oval)
- `Name` → simple attribute (unique per requirements, but Number is the diagrammatic key)
- `Location` → simple, single-valued attribute —*"a single location "*confirms no double oval needed

> **Assumption — WORKS_FOR (Department side)**
>
> **Stated:** Every employee must belong to exactly one department → Employee side is **total**.
>
> **Assumption:** The diagram shows a **single line** on the Department side, meaning a department can exist with zero employees assigned to it → Department side is **partial**, min = 0, max = N.
>
> *This is a design choice. You could require every department to have at least one employee by making their side total (double line).*

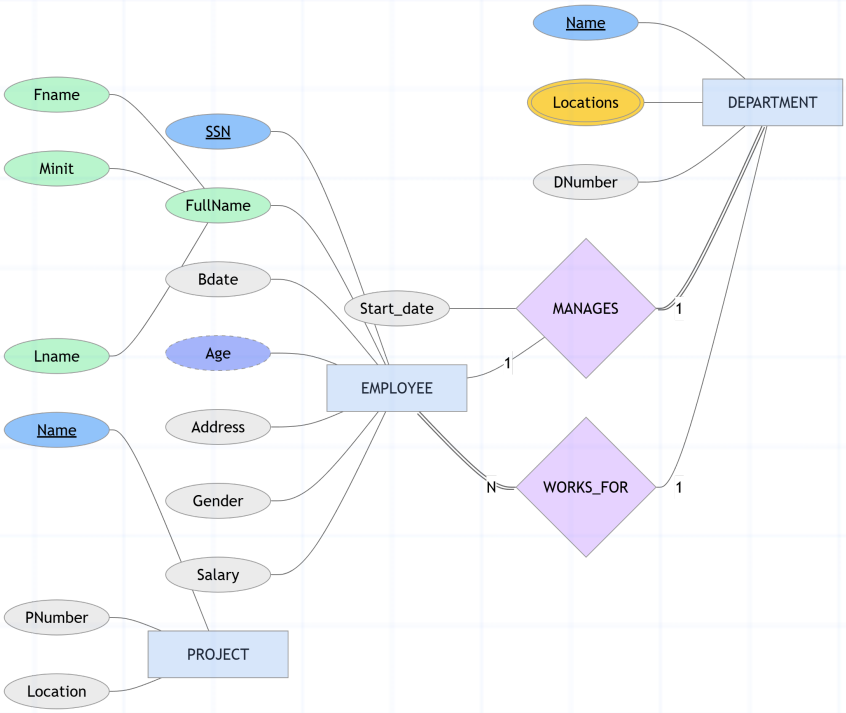


Diagram 21: Step 4 — PROJECT Entity

4.3.6 Step 5 — CONTROLS Relationship

> **From Requirements**
>
> Each project is controlled by exactly one department.

CONTROLS connects PROJECT and DEPARTMENT with cardinality N:1.

- **Project side:** *"exactly one"*→ **total** participation (double line), max = 1
- **Department side:** The requirement says nothing about whether a department must control any projects.

> **Assumption — CONTROLS (Department side)**
>
> **Stated:** Every project is controlled by exactly one department → Project side is **total**.
>
> **Assumption:** A department may control zero or many projects → Department side is **partial**, min = 0, max = N.
>
> *The requirements do not require every department to control a project.*

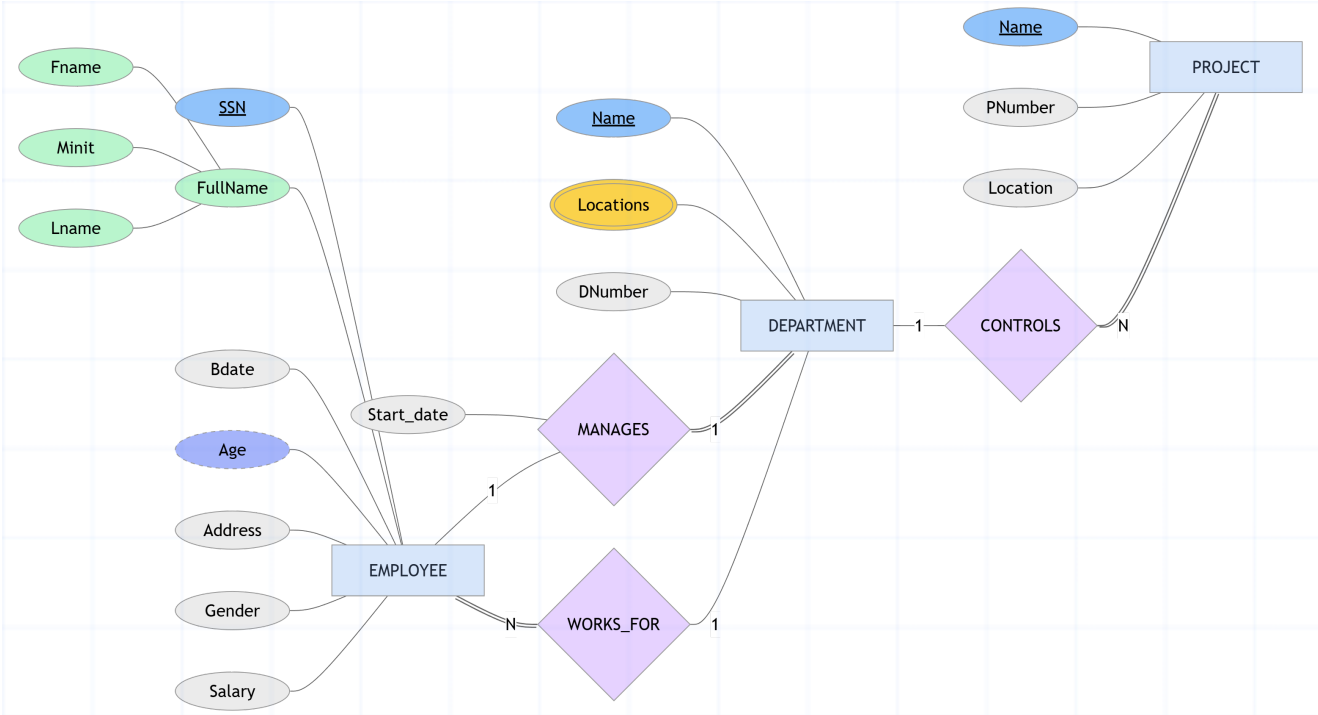


Diagram 22: Step 5 — CONTROLS Relationship

4.3.7 Step 6 — WORKS_ON Relationship

> **From Requirements**
>
> *Each employee **may** work on several **projects**. We track the **number of hours** per week each employee works on each project.*

WORKS_ON connects EMPLOYEE and PROJECT with cardinality M:N.

- `Hours` → **relationship attribute** on WORKS_ON — it belongs to the (Employee, Project) pairing, not to either entity alone
- **Employee side:** The word **"may"** explicitly makes this **partial**(min = 0), max = N
- **Project side:** Not stated — see assumption below.

> **Assumption — WORKS_ON (Project side)**
>
> **Stated:** *"May work on"*→ Employee participation is **partial**— directly from requirements.
>
> **Assumption:** A project that has no employees assigned is not yet operational → Project side is **total**, min = 1, max = N.
>
> *This assumption must be confirmed with stakeholders — it is not stated in the text.*

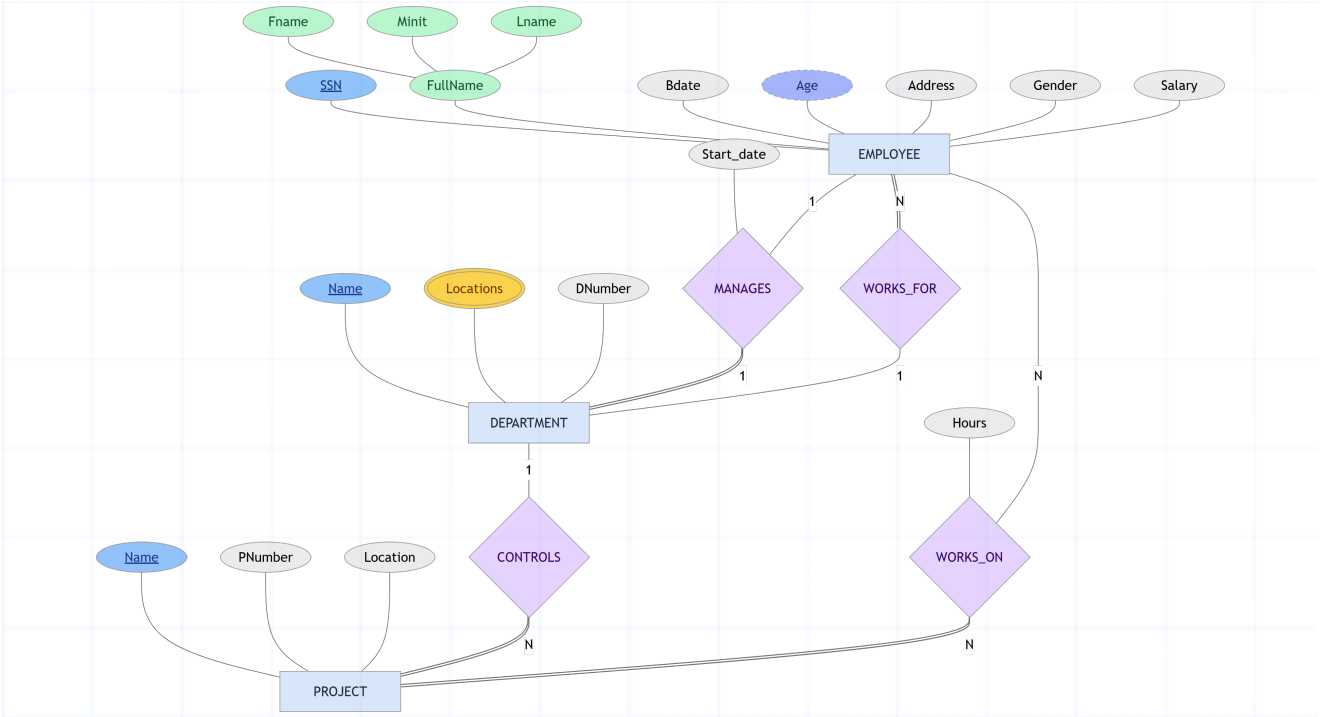


Diagram 23: Step 6 — WORKS_ON Relationship

4.3.8 Step 7 — SUPERVISION Relationship

> **From Requirements**
>
> *Each employee may be supervised by at most one other employee.*

SUPERVISION is a recursive (unary) relationship — EMPLOYEE related to itself, with two roles: Supervisor and Supervisee.

- **Supervisee side:** *"may be supervised"* → **partial**(min = 0);*"at most one"*→ max = 1
- **Supervisor side:** Not stated — see assumption below.
- **Project side:** Not stated — see assumption below.

> **Assumption — SUPERVISION (Supervisor side)**
>
> **Stated:** Each employee *may* be supervised by at most one → Supervisee side is **partial, max = 1**.
>
> **Assumption:** Not every employee supervises anyone (e.g., a regular worker has no one under them; the top-level manager also has no supervisor) → Supervisor side is **partial**, min = 0, max = N.

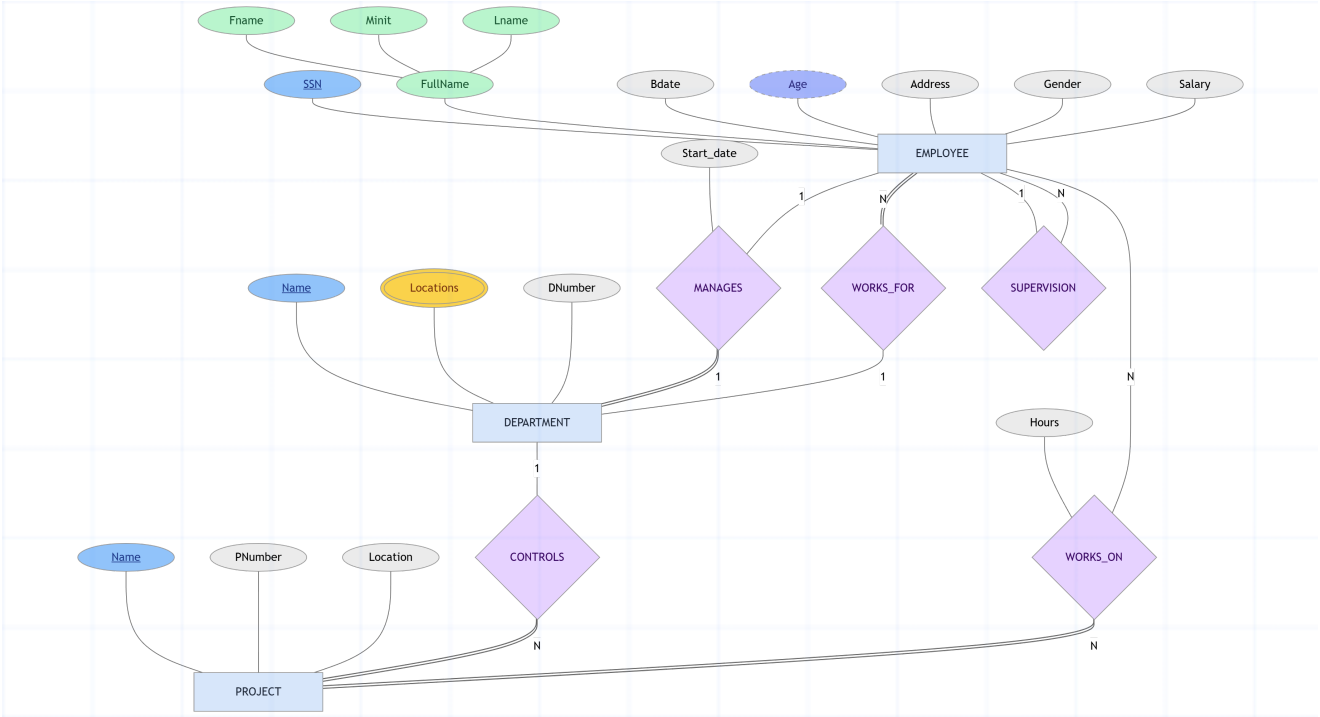


Diagram 24: Step 7 — SUPERVISION Relationship

4.3.9 Step 8 — DEPENDENT Weak Entity + HAS (Full Diagram)

> **From Requirements**
>
> *Employees may have dependents. Each dependent has a name, gender, date of birth, and relationship to the employee.*

DEPENDENT is a weak entity — it has no key of its own and cannot exist without its owning employee.

- `Name` → **partial key**(dashed-underline oval) — unique within one employee's dependents, not globally
- `Gender` , `Bdate` , `Relationship` → simple attributes

- `HAS` is the **identifying relationship**(double diamond) — the combination `(SSN, Name)` uniquely identifies each dependent
- **Employee side:** *"may have"* →**partial**(min = 0, max = N)
- **Dependent side:** A dependent cannot exist without an employee →**total**(min = 1) — this follows structurally from the weak entity definition, not from an assumption

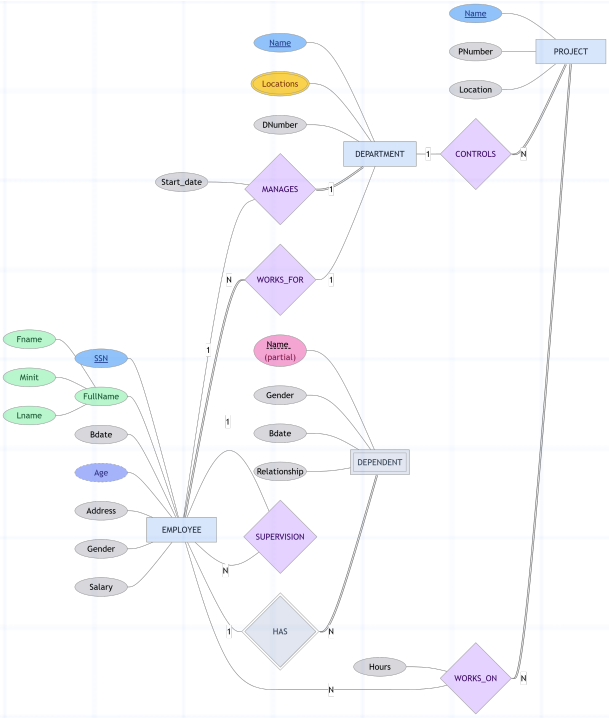


Diagram 25: Step 8 — DEPENDENT Weak Entity + HAS (Full Diagram)

5 Min-Max Notation

Chen notation tells us *whether* a participation is total or partial and *what* the cardinality ratio is (1:1, 1:N, M:N), but it stops short of answering a frequent business question: *exactly how many*? “At most three active orders,” “between two and five team members,” “exactly one department chair” — these rules all collapse into Chen’s coarse labels. **Min-Max notation** was introduced precisely to close that gap: it keeps the same diagram shape but replaces each line-end label with a numeric pair (min, max) that pins down both bounds explicitly.

5.1 What Do min and max Mean?

- **min**— the minimum number of relationship instances this entity *must* participate in:
  - min=0→ participation is **optional**(equivalent to a single line in Chen)
  - min≥1→ participation is **mandatory**(equivalent to a double line in Chen)
- **max**— the maximum number of relationship instances this entity *can* participate in:
  - max=1→ at most one
  - max=N(or∞) → no upper limit

(min,max) pair	English reading	Chen equivalent
(0,1)	“zero or one”	Partial, cardinality 1
(1,1)	“exactly one”	Total, cardinality 1
(0,N)	“zero or more”	Partial, cardinality N
(1,N)	“one or more”	Total, cardinality N
(4,N)	“at least four”	Total, cardinality N

Why use Min-Max instead of Chen labels?

Chen notation labels each relationship line with 1, N, or M — this captures the *maximum* cardinality but not the *minimum*. Min-Max adds the minimum, making it immediately clear whether participation is optional or mandatory without relying on single-vs-double lines. It is especially useful when writing precise business rules for a development team to implement.

5.2 Placement Rule

The (min, max) pair is written on the line **next to the entity it constrains**. The general form is:

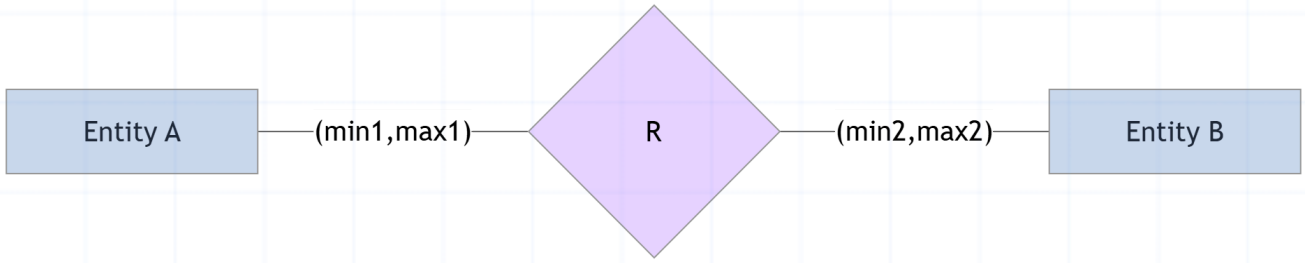


Diagram 26: General form of Min-Max placement on a binary relationship.

- An instance of Entity A may be related to at least min1 and at most max1 instances of Entity B.
- Likewise, min2 is the minimum number and max2 is the maximum number of instances of Entity A in to which Entity B is related.

EXAMPLE 17: **Min-Max notation on an EMPLOYEE–DEPARTMENT relationship (illustrative values).**

The diagram below shows Min-Max annotation on a generic EMPLOYEE–DEPARTMENT relationship. The values here are **chosen to illustrate specific constraint patterns** and are independent of the Company Database worked example.

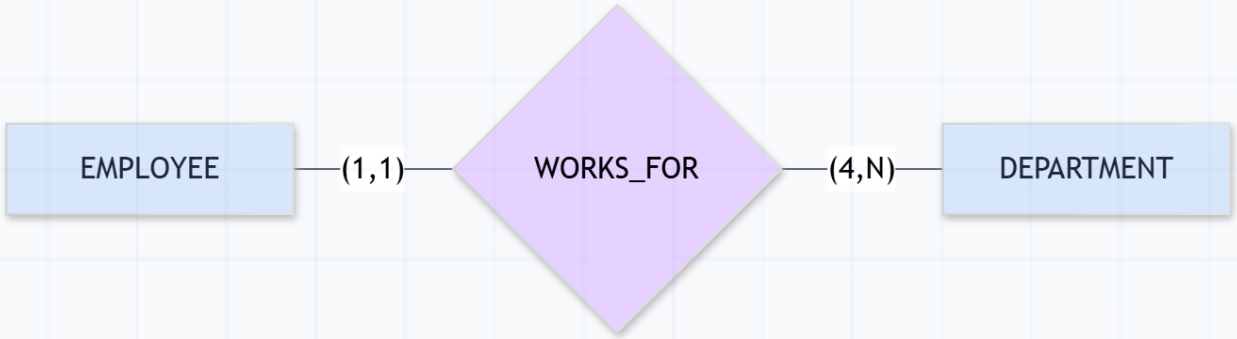


Diagram 27: Min-Max notation on an EMPLOYEE–DEPARTMENT relationship (illustrative values).

Pair	Placed next to	Meaning
(1,1)	EMPLOYEE	Every employee is associated with exactly one department — mandatory and singular.
(4,N)	DEPARTMENT	Every department has at least 4 employees — a specific minimum business rule.

5.3 Company Database in Min-Max Notation

The table below re-expresses every relationship from the Worked Example using (min,max) pairs, consistent with the Full Constraint Summary and the Chen diagram.

Relationship	Entity A	(min,max) on A	Entity B	(min,max) on B
WORKS_FOR	EMPLOYEE	(1,1)	DEPARTMENT	(0,N)
MANAGES	EMPLOYEE	(0,1)	DEPARTMENT	(1,1)
WORKS_ON	EMPLOYEE	(0,N)	PROJECT	(1,N)
CONTROLS	PROJECT	(1,1)	DEPARTMENT	(0,N)
SUPERVISION	EMPLOYEE (Supervisee)	(0,1)	EMPLOYEE (Supervisor)	(0,N)
HAS	EMPLOYEE	(0,N)	DEPENDENT	(1,1)

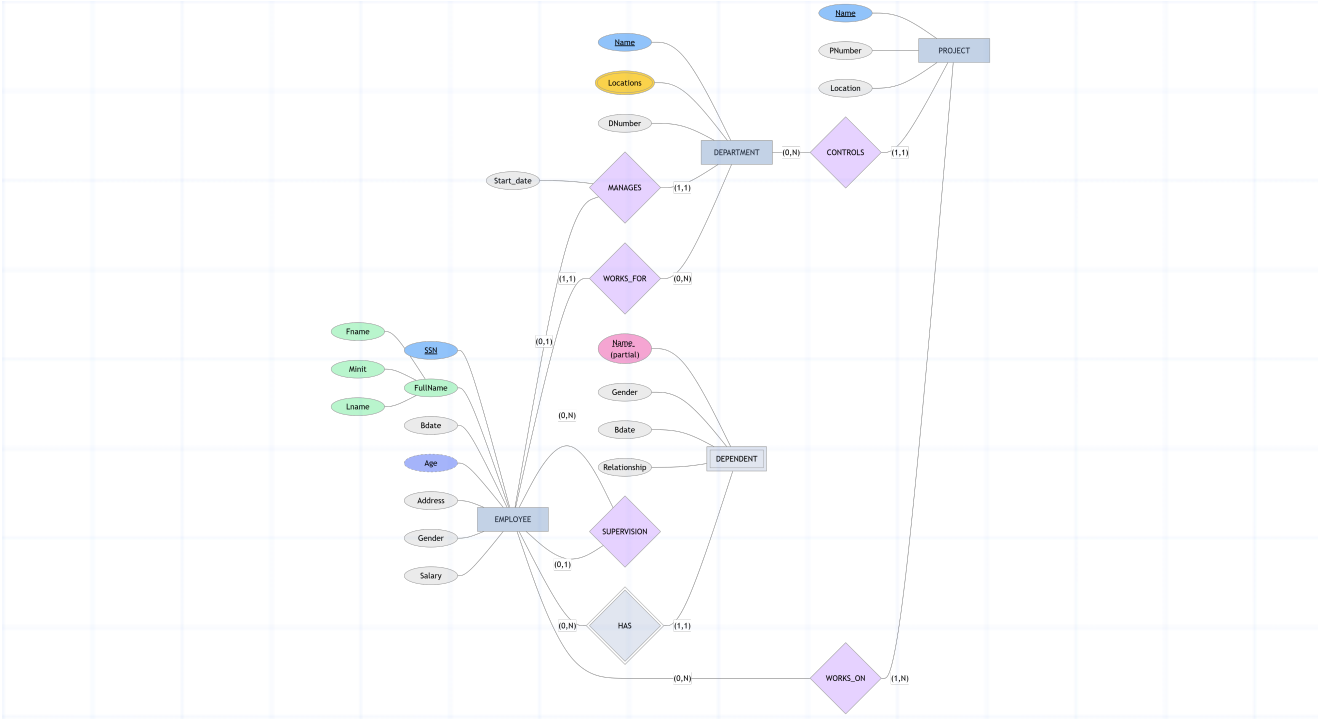


Diagram 28: Full Company ER diagram annotated with Min-Max notation.

5.4 Chen vs Min-Max: Side-by-Side Examples

EXAMPLE 18: Professor CHAIRS Department

The university has several departments. Each department must have exactly one chairperson who must be a professor. A professor can chair at most one department. Some professors may not chair any department — the chairperson role is optional for professors. If a professor is appointed as chair, they must chair exactly one department. No department can exist without a chairperson.

> **Note**
>
> A department chair is*a faculty member in a given academic department*, assigned by the dean to manage that department.

Show Solution

Relationship: CHAIRS

Property	Entity A — Professor	Entity B — Department
Participation	Partial (optional)	Total (mandatory)
(min,max)	(0,1)	(1,1)
Chen cardinality label	1	1

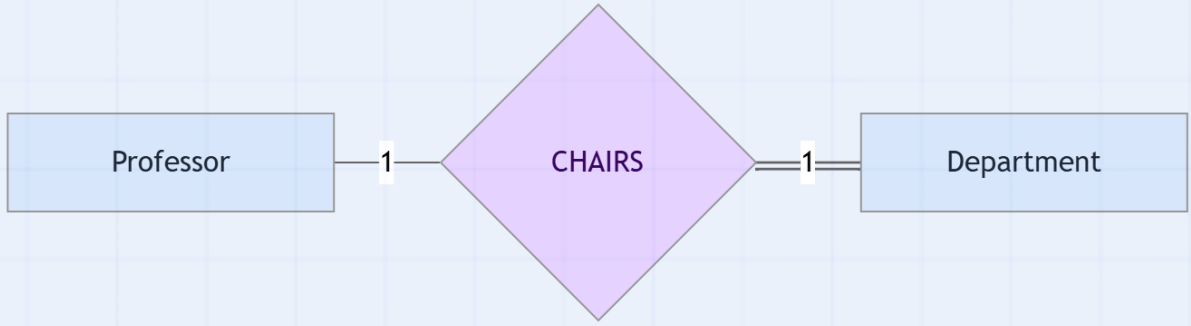


Diagram 29: Professor CHAIRS Department (Chen)

How to read — Chen:

- **Left → Right (LTR):** Each Professor chairs **0 or 1** Department.
- **Right → Left (RTL):** Each Department is chaired by **exactly 1** Professor.
- The **double line** on the Department side means Department participation is **total** (mandatory).
- The **single line** on the Professor side means Professor participation is **partial** (optional).

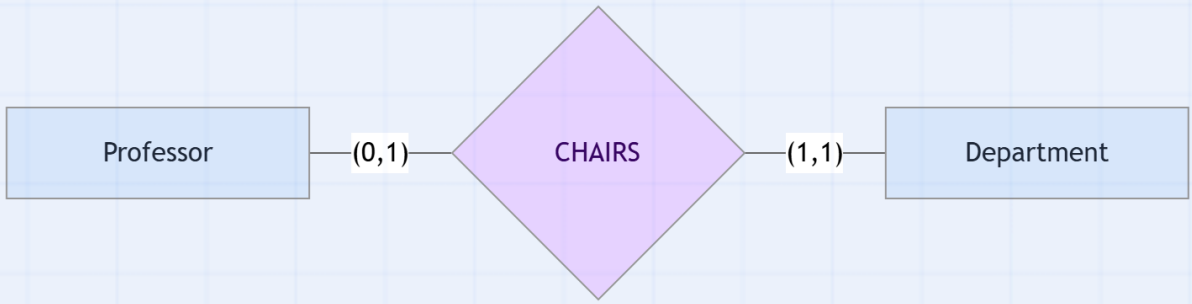


Diagram 30: Professor CHAIRS Department (Min-Max)

How to read — Min-Max:

- A professor chairs **at most one** department — but is not required to chair any.
- Every department has **exactly one** chair — mandatory, and only one.

> **Key insight**
>
> Chen told us the *ratio* (1:1) but left participation implicit in the line style. Min-Max spells out both the minimum and maximum in a single, self-documenting pair.

EXAMPLE 19: Student HOLDS_PERMIT Parking_Permit

Students may obtain a parking permit from the university. A student can have at most one parking permit. Some students don't have cars and therefore have no permit. Each parking permit is assigned to exactly one student, but permits can exist in the system before being assigned. If a student leaves, their permit becomes available for reassignment.

Show Solution

Relationship: HOLDS_PERMIT

Property	Entity A — Student	Entity B — Parking_Permit
Participation	Partial (optional)	Partial (optional)
(min,max)	(0,1)	(0,1)
Chen cardinality label	1	1

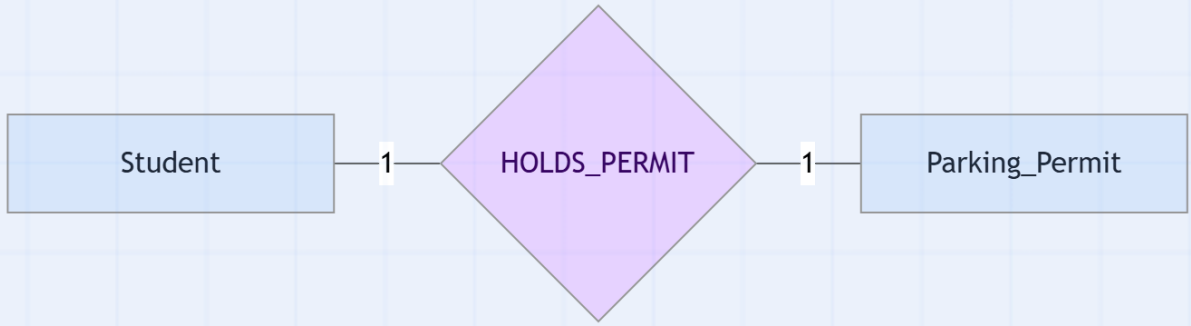


Diagram 31: Student HOLDS_PERMIT Parking_Permit (Chen)

How to read — Chen:

- **Left → Right (LTR):** Each Student holds **0 or 1** Parking Permit.
- **Right → Left (RTL):** Each Parking Permit is held by **0 or 1** Student.
- **Single line** on both sides → participation is **partial** on both ends (neither student nor permit is required to be matched).

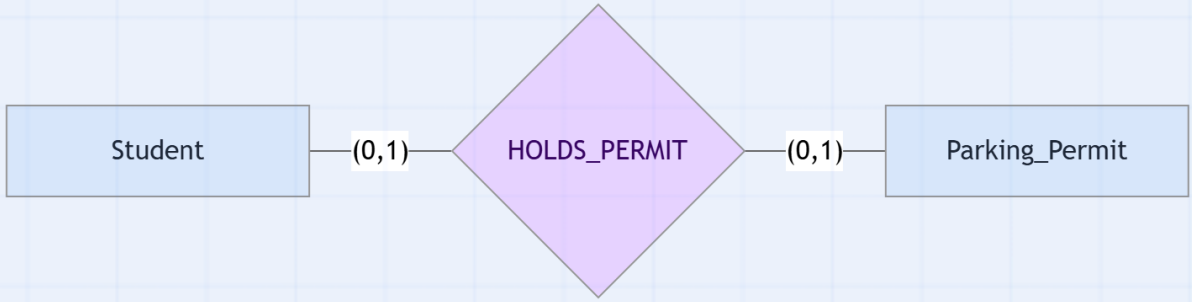


Diagram 32: Student HOLDS_PERMIT Parking_Permit (Min-Max)

How to read — Min-Max:

- A student holds **at most one** parking permit — but may hold none.
- A parking permit is held by **at most one** student — but may be unassigned.

> **Key insight**
>
> Both sides are optional. Min-Max makes this unambiguous without having to check whether lines are single or double.

> **Note**
>
> The full implementation of reassignment — `ON DELETE SET NULL`, status trigger, and the design discussion `SET NULL` vs. `CASCADE`

EXAMPLE 20: Department OFFERS Course

Every department in the university must offer at least one course. A course must be offered by exactly one department. Departments can offer multiple courses. Courses cannot exist without being associated with a department. New departments must create at least one course within their first semester.

✅ Show Solution

Relationship: OFFERS

Property	Entity A — Department	Entity B — Course
Participation	Total (mandatory)	Total (mandatory)
(min,max)	(1,N)	(1,1)
Chen cardinality label	N	1



Diagram 33: Department OFFERS Course (Chen)

How to read — Chen:

- **Left → Right (LTR):** Each Department offers **1 or more** Courses.
- **Right → Left (RTL):** Each Course is offered by **exactly 1** Department.
- **Double line** on both sides → participation is **total** on both ends.



Diagram 34: Department OFFERS Course (Min-Max)

How to read — Min-Max:

- Every department offers **at least one** course (no upper limit).
- Every course is offered by **exactly one** department.

> **Key insight**
>
> Chen's double lines on both sides simply said "mandatory on both ends." Min-Max adds the upper bound: a department can offer *many* courses, but a course belongs to *only one* department.

> **Warning**
>
> The static ER diagram cannot capture the *timing* rule "within the first semester." The enforcement trigger

EXAMPLE 21: Professor ADVISES Student

Professors may serve as academic advisors to students. A professor can advise multiple students. Some professors choose not to advise any students. A student may have at most one academic advisor. Some students, particularly freshmen, may not have an assigned advisor yet. Each advisor-student relationship is formalized through the advising system.

Show Solution

Relationship: ADVISES

Property	Entity A — Professor	Entity B — Student
Participation	Partial (optional)	Partial (optional)
(min,max)	(0,N)	(0,1)
Chen cardinality label	N	1

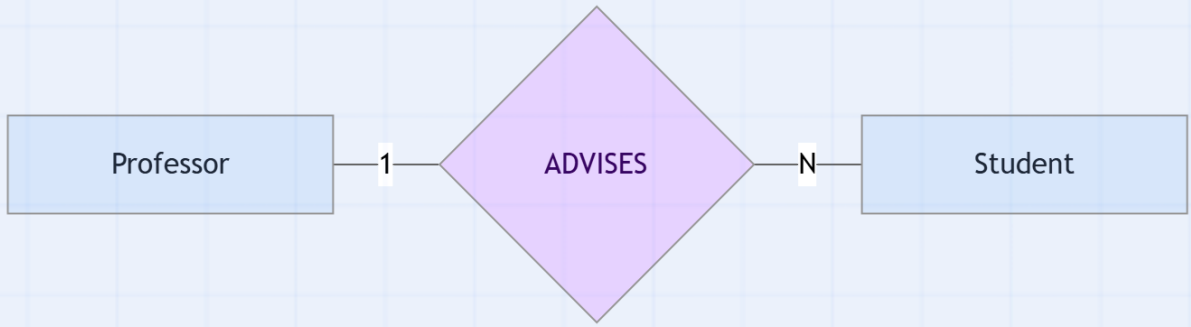


Diagram 35: Professor ADVISES Student (Chen)

How to read — Chen:

- **Left → Right (LTR):** Each Professor advises **0 or more** Students.
- **Right → Left (RTL):** Each Student is advised by **0 or 1** Professor.

- ◦ **Single line** on both sides → participation is **partial** on both ends.

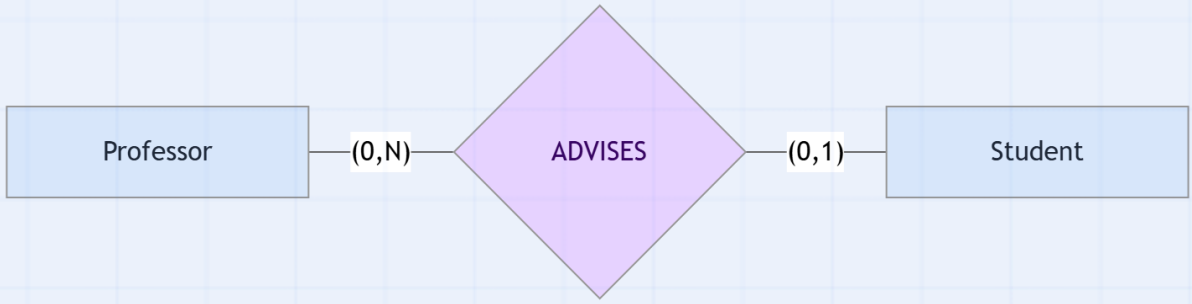


Diagram 36: Professor ADVISES Student (Min-Max)

How to read — Min-Max:

- A professor may advise **any number** of students — or none at all.
- A student may have **at most one** advisor — but is not required to have one.

> **Key insight**
>
> Chen indicated a 1:N ratio with optional participation, but gave no upper bound on the N side. Min-Max confirms there truly is no cap on how many students a professor can advise, while also confirming a student cannot have more than one.

The phrase *"Each advisor-student relationship is formalized through the advising system"* is a strong hint that **ADVISES is not a simple binary connection** — it carries descriptive information of its own. In ER modelling, when a relationship has data associated with it (not belonging to either entity alone), that data is modelled as **relationship attributes**.

Attribute	Type	Meaning
assignment_date	DATE	When the advising relationship was established
status	ENUM(ACTIVE, INACTIVE, PENDING)	Current state of the advisory relationship
end_date	DATE (nullable)	When the relationship was formally terminated
notes	TEXT	Free-text notes entered by advisor or registrar

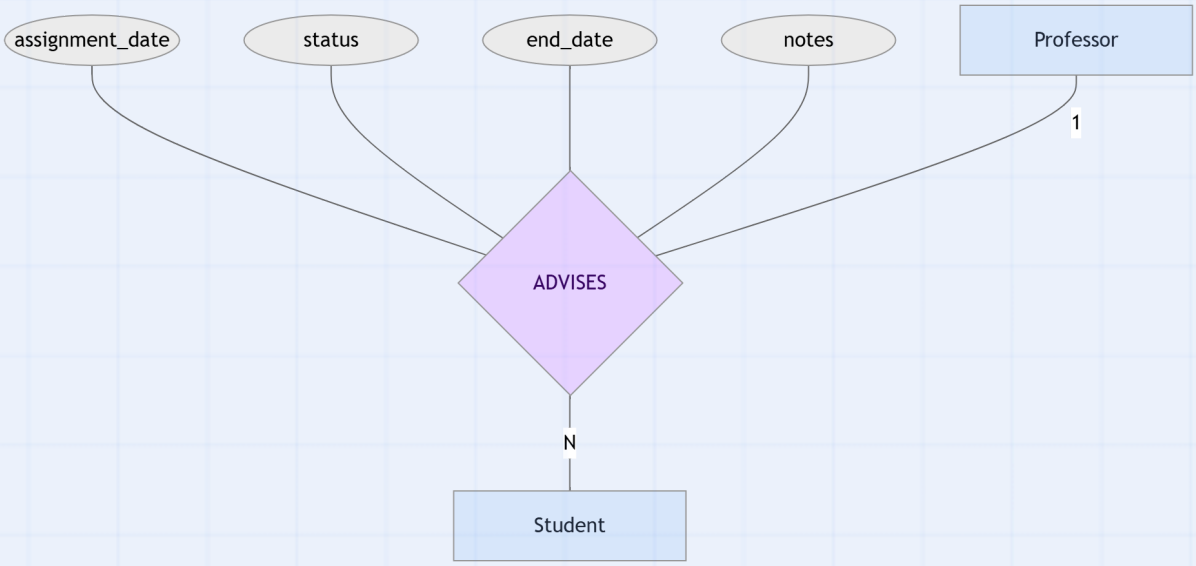


Diagram 37: ADVISES relationship with attributes (assignment_date, status, end_date).

5.5 Specific Value Constraints: (0 , 3) (0,3), (2 , 5) (2,5), and Beyond

> **Not every business rule fits neatly into the four standard patterns**
> **(0,1), (1,1), (0,N), (1,N)**
>
> Min-Max notation allows **any non-negative integers** as the min and max values.

Example	English meaning	Typical business rule
(0,3)	zero to three	A customer may place at most 3 active orders at a time
(2,5)	two to five	A project team must have between 2 and 5 members
(1,4)	one to four	An employee manages at least 1, at most 4 direct reports
(3,3)	exactly three	A committee seat is always filled by exactly 3 co-chairs

These constraints are **perfectly valid** at the conceptual level — they are precise expressions of real business rules that the common patterns simply cannot capture. Write them on your ER diagram exactly as you would any other (min,max) pair.

> ⚠ **Conceptual validity ≠ automatic SQL enforcement**
>
> Standard SQL foreign-key constraints only know how to enforce "must exist" (referential integrity). They have no mechanism to *count* how many rows a related entity participates in before accepting or rejecting a change. Specific value constraints such as (0,3) or (2,5) therefore **cannot be enforced by foreign keys alone**.
>
> Enforcing specific value constraints in the database requires either subquery-`CHECK`s (PostgreSQL), triggers, stored procedures, or application logic.

6 Enhanced Entity-Relationship (EER) Modeling

6.1 The Scenario: Hospital Staff Management (AAUP Medical Center)

At the **Arab American University Medical Center (AAUP Center)** in Jenin, database designers need to model various employee types (`DOCTOR` ,

`RESEARCHER` , `NURSE` , `SECRETARY` , `TECHNICIAN` , `ENGINEER`).

All staff members share foundational attributes: `SSN` , `Name` , `Address` , and `Salary` . However, specific roles require unique local attributes (e.g., `Specialty` for doctors, `EquipmentCert` for technicians). Furthermore, employees are categorized by payment method (`SALARIED_EMPLOYEE` vs. `HOURLY_EMPLOYEE`) and leadership roles (`MANAGER`).

This domain presents complex real-world business constraints:

- **Strict Disjointness:** An employee must be **either** Salaried **or** Hourly.
- **Overlapping Roles:** A single individual can hold multiple roles simultaneously. For example, **Dr. Tariq Zaid** serves as an attending physician in Cardiology (`DOCTOR`) while also directing cardiovascular research trials (`RESEARCHER`), or acting as the department head (`MANAGER`).

6.1.1 Case 1: Separate Entities (Duplicated Attributes)

In a standard ER model without superclass/subclass features, each functional role is modeled as a completely independent entity table (`DOCTOR` , `RESEARCHER` , `TECHNICIAN` , etc.). Base attributes (`SSN` , `Name` , `Address` , `Salary`) are repeated inside every single table.

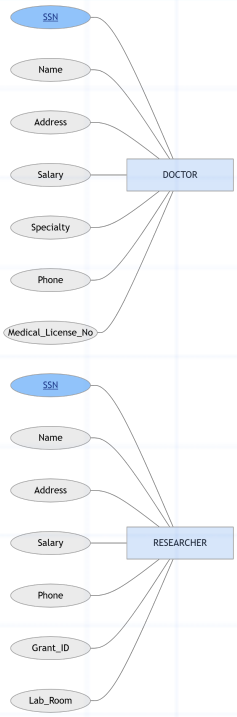


Diagram 38: Standard ER Diagram for Case 1: Independent Subclass Entities (Duplicated Base Attributes)

Issues & Vulnerabilities in Case 1:

- **Update Anomalies:** If Dr. Tariq moves from Jenin to Nablus, an `UPDATE` statement executed on the `DOCTOR` table leaves his record in the `RESEARCHER` table out of date, resulting in an inconsistent database state.
- **Complex Global Queries:** Retrieving a roster of all hospital personnel requires executing a costly `UNION` across 5+ tables (`SELECT ... FROM DOCTOR UNION SELECT ... FROM RESEARCHER ...`).
- **High Maintenance Overhead:** Modifying a shared property (e.g., increasing the character limit of `Address`) requires running schema migrations (`ALTER TABLE`) on every role table individually.
- **Relationship Clutter:** Connecting employees to another entity (e.g., `DEPARTMENT`) requires drawing separate, redundant relationship links for every entity type (`DOCTOR_WORKS_IN` , `NURSE_WORKS_IN` , etc.).

6.1.2 Case 2: Single Table with All Attributes

To avoid separate tables, another standard ER approach collapses every role into a single, massive `EMPLOYEE` table containing all shared attributes along with every role-specific attribute (`Specialty` , `EquipmentCert` , `TypingSpeed` , `Grant_ID` , `PayScale`).

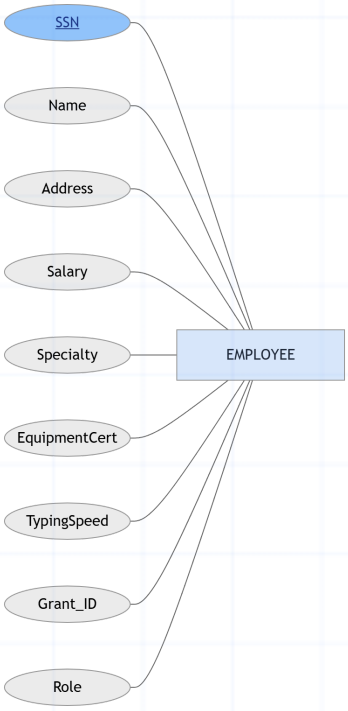


Diagram 39: Standard ER Diagram for Case 2: Single Table with All Attributes

SSN (PK)	Name	Address	Salary	Specialty	EquipmentCert	TypingSpeed	Grant_ID	Role
401928	Tariq Zaid	Jenin	2500	Cardiology	NULL	NULL	AAUP-202	Doc/Res
302194	Ahmad Omar	Nablus	1200	NULL	MRI Specialist	NULL	NULL	Tech

Issues & Vulnerabilities in Case 2:

- **Massive `NULL` Value Overhead:** Because most employees fill only one or two specific roles, most role-specific columns contain `NULL` values, wasting storage and diluting data density.
- **Inability to Enforce Mandatory Attributes:** If a doctor *must* have a `Specialty` , you cannot define `Specialty` as `NOT NULL` at the database level because non-doctor rows (nurses, secretaries) would fail validation.
- **Loss of Semantic Clarity:** Combining distinct conceptual entities into one flat table hides business logic and forces reliance on artificial type-discriminator flags (e.g., `Job_Type` strings).

6.1.3 The Need for Enhanced ER (EER) Modeling

Standard ER modeling lacks the constructs needed to represent complex organizational domains efficiently. Enhanced ER introduces specialized concepts that solve the defects of both Case 1 and Case 2 by incorporating semantic and object-oriented concepts like **inheritance, specialization, and generalization**.

EER Addresses Complex Domains:

1. **Attribute Inheritance:** Subtype entities automatically inherit all attributes and relationships of their supertype entity, eliminating redundancy.
2. **Specialization (Top-Down):** The process of maximizing differences by identifying lower-level subgroups from a higher-level entity.
3. **Generalization (Bottom-Up):** The process of minimizing differences by identifying a higher-level generalized entity from multiple lower-level subgroups.

6.2 Subclasses and Superclasses in EER Diagrams

In Enhanced Entity-Relationship (EER) modeling, **superclass/subclass relationships** allow you to represent specialization and generalization, similar to inheritance in object-oriented programming.

📖 DEFINITION 20: **Superclass**

An entity type that includes distinct subgroupings of its entities that require explicit representation (e.g., `EMPLOYEE`).

📖 DEFINITION 21: **Subclass**

A subgrouping of entities from a superclass that shares common attributes and relationships with the superclass, but may have additional specific attributes or relationships of its own (e.g., `SECRETARY`, `ENGINEER`, `MANAGER`).

📝 EXAMPLE 22:

`EMPLOYEE` may be further grouped into:

- `SECRETARY`, `ENGINEER`, `TECHNICIAN`, ... : Based on the **EMPLOYEE's Job**.
- `MANAGER`: EMPLOYEEs who are managers.
- `SALARIED_EMPLOYEE`, `HOURLY_EMPLOYEE`: Based on the **EMPLOYEE's method of pay**.

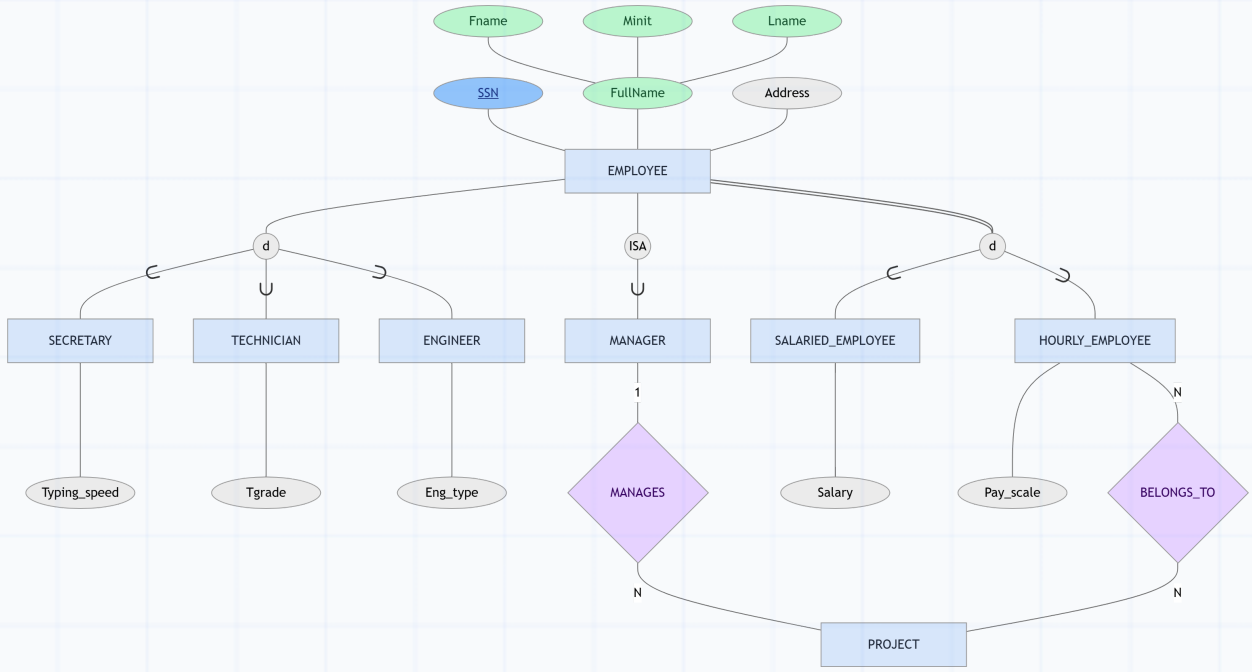


Diagram 40: EER diagram notation illustrating superclass/subclass specialization constraints (disjointness, overlapping, and single-subclass IS-A relationships) for an EMPLOYEE entity.

Note

- EMPLOYEE is the superclass for the three specializations: SECRETARY, ENGINEER, TECHNICIAN}, {MANAGER}, SALARIED_EMPLOYEE, HOURLY_EMPLOYEE}.
- Each of these subgroupings is a subset of EMPLOYEE entities called a subclass of EMPLOYEE.
- These are called IS-A relationships: SECRETARY IS-A EMPLOYEE, TECHNICIAN IS-A EMPLOYEE, ...
- An entity cannot exist in the database merely by being a member of a subclass; it must also be a member of the superclass.
- A member of the superclass can be optionally included as a ember of any number of its subclasses.
- A salaried employee who is also an engineer belongs to the two subclasses: ENGINEER, and SALARIED_EMPLOYEE.
- A salaried employee who is also an engineering manager belongs to the three subclasses: MANAGER, ENGINEER, and ALARIED_EMPLOYEE.
- It is not necessary that every entity in a superclass be a member of some subclass.
- An entity that is member of a subclass inherits All attributes of the entity as a member of the superclass and All relationships of the entity as a member of the superclass.

6.3 Generalization

DEFINITION 22: **Generalization**

Generalization is a bottom-up design process that serves as the inverse of specialization. It combines multiple lower-level entity types that share common characteristics into a single higher-level superclass.

EXAMPLE 23: CAR, TRUCK generalized into VEHICLE

Designing independently, you create `CAR` and `TRUCK` all with `LicensePlate`, `Make`, `Model`, `Year`. Noticing the overlap, you generalize them into `VEHICLE`, extracting the shared attributes:

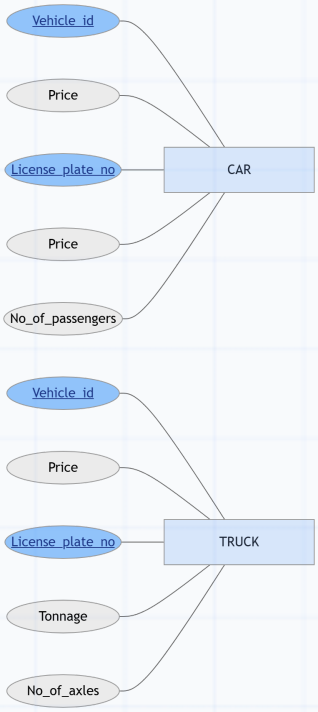


Diagram 41: CAR and TRUCK

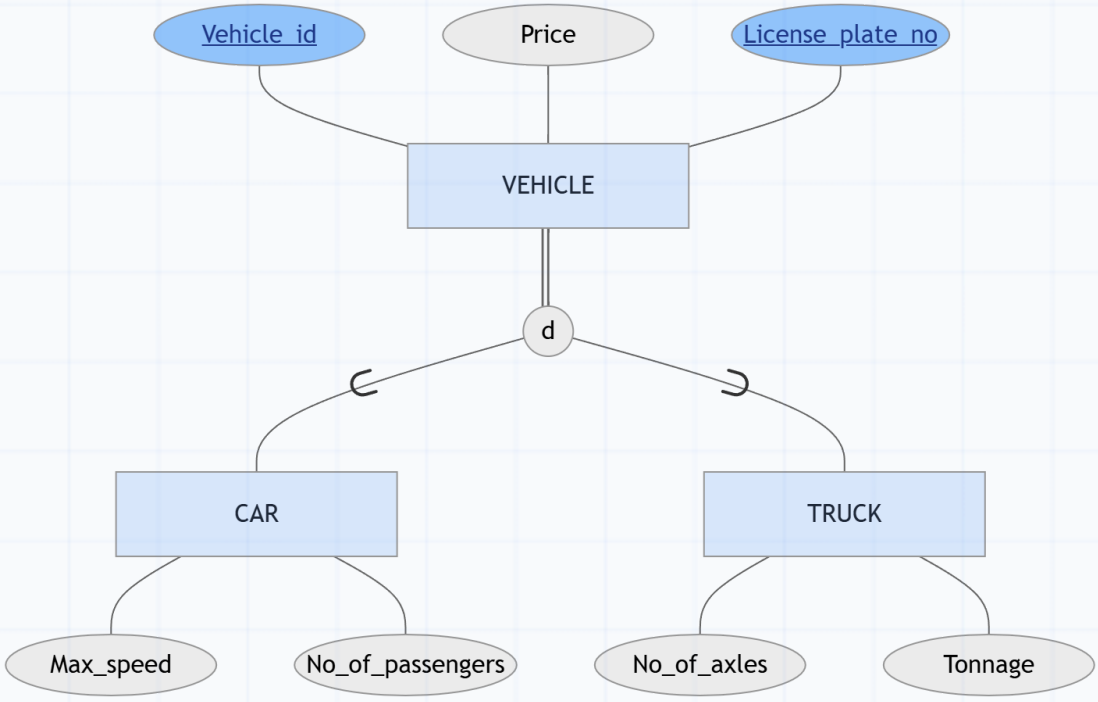


Diagram 42: Generalizing CAR and TRUCK into the superclass VEHICLE

Specialization vs. Generalization

Specialization vs. Generalization produce identical EER diagrams. The distinction is purely in the *decision process* (top-down design vs. bottom-up abstraction). In practice, real designs mix both: you may generalize two existing entities and then specialize the resulting superclass further.

6.4 Specialization

📖 DEFINITION 23: **Specialization**

Specialization is a top-down design process in Enhanced Entity-Relationship (EER) modeling where a higher-level entity type (superclass) is divided into lower-level entity types (subclasses) based on distinguishing characteristics or job roles.

📝 EXAMPLE 24: **University System — PERSON**

Superclass: `PERSON`

- *Common attributes:* `Person_ID`, `Name`, `Email`, `Phone_Number`

Subclasses (Specializations):

- `STUDENT`: Has specific attributes like `GPA`, `Major`, `Credits_Completed`.
- `PROFESSOR`: Has specific attributes like `Office_Hours`, `Department`, `Tenure_Status`, `Salary`.

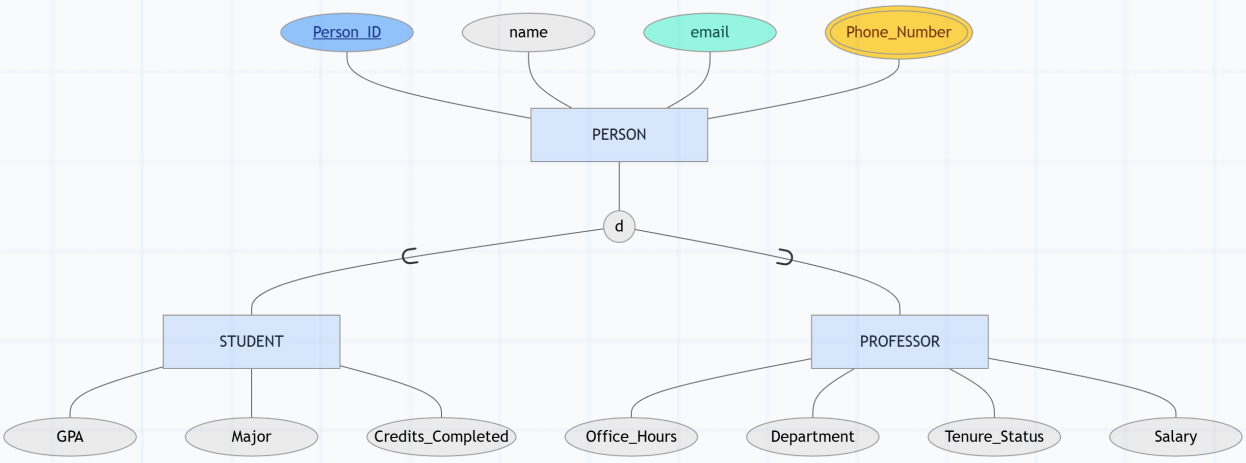


Diagram 43: University System — PERSON (Specializations)

Why specialize?

A standard `PERSON` does not have a `GPA` or `Tenure_Status`. Adding these specific fields to a single table would result in a lot of empty (NULL) values for people who aren't students or professors.

6.5 Constraints on Specialization

A specialization (or generalization) hierarchy is governed by two independent sets of constraints that dictate how instances flow from the superclass into the subclasses.

6.5.1 Disjointness Constraint

The disjointness constraint determines whether an entity instance can belong to **more than one** subclass simultaneously within the same specialization set.

- **Disjoint (d):** Each entity instance can belong to **at most one** of the subclasses. The subclasses are mutually exclusive.
  - **Example:** In an EMPLOYEE specialization, an employee can be either an `HOURLY_EMPLOYEE` or a `SALARIED_EMPLOYEE` —never both at the same time.
- **Overlapping (o):** An entity instance **may belong to multiple subclasses** simultaneously.
  - **Example:** At a university, a `PERSON` can be both a `STUDENT` and an `EMPLOYEE` (such as a graduate teaching assistant) at the same time.

📝 EXAMPLE 25: **The Faculty Ph.D. Candidate**

Imagine a university database for **Arab American University** that models people on campus.

1. **The Core Entities:**
   - `PERSON` (Superclass): Stores general demographic details like `Person_ID`, `First_Name`, `Last_Name`, and `Email`.
   - `INSTRUCTOR` (Subclass): Stores faculty-specific attributes like `Office_Number`, `Rank`, and `Salary`.
   - `STUDENT` (Subclass): Stores student-specific attributes like `Student_ID`, `Major`, and `Cumulative_GPA`.
2. **The Real-World Situation:**
   - Meet **Dr. Samer**, who is a full-time faculty member in the Computer Systems Engineering department teaching undergraduate courses.
   - At the same time, Dr. Samer is enrolled as a part-time doctoral researcher in a Ph.D. program, meaning he takes advanced coursework and registers for dissertation credits.

✅ Show Solution

Why Do We Need the Overlapping (o) Constraint Here?

If the specialization of `PERSON` into `INSTRUCTOR` and `STUDENT` were defined as **Disjoint (d)** (mutually exclusive):

- The database schema would enforce a strict rule: **An entity instance can belong to *at most one* subclass.**
- When Dr. Samer is added to the system, the database forces a choice: Is he *only* an `INSTRUCTOR` or *only* a `STUDENT`?
- **The Problem:** If you classify him strictly as an `INSTRUCTOR`, the system drops his `Student_ID`, `GPA`, and course enrollment records. If you classify him strictly as a `STUDENT`, the system strips away his `Office_Number`, teaching

assignments, and faculty salary profile. **You lose critical real-world data.**

By setting the constraint to **Overlapping (*o*)**, the EER model allows Dr. Samer's single `PERSON` instance to simultaneously link to **both** subclass tables (`INSTRUCTOR` and `STUDENT`):

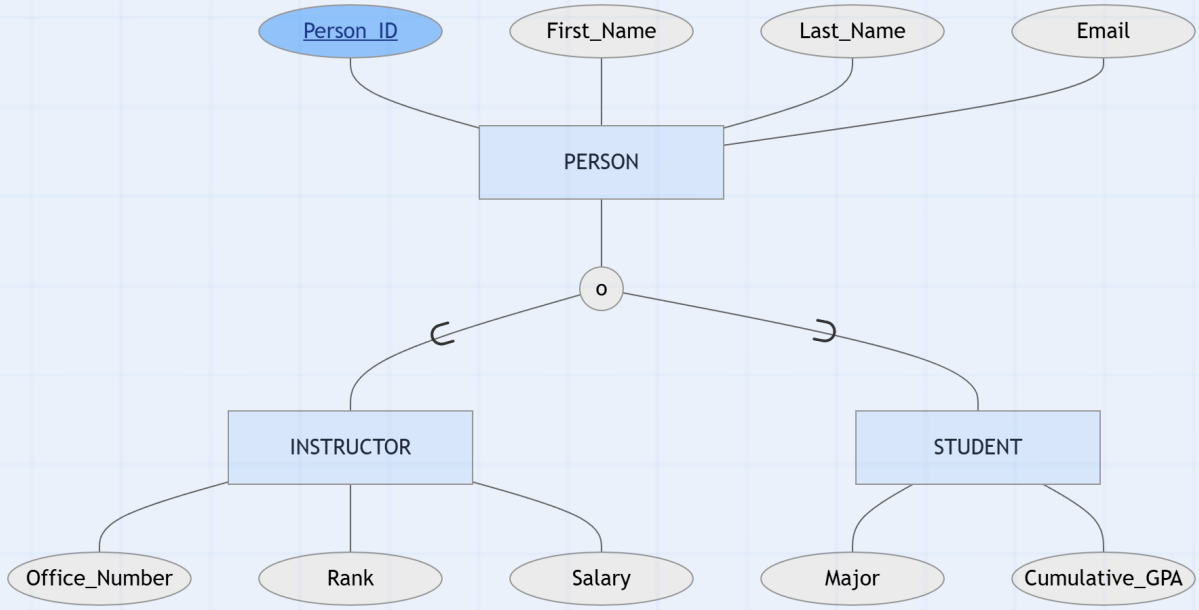


Diagram 44: The Faculty Ph.D. Candidate

6.5.2 Completeness Constraint

The completeness constraint determines whether every instance in the superclass *must* participate in at least one subclass, or if it can exist purely as a superclass instance.

- **Total:** Represented visually by a **double line** connecting the superclass to the circle. Every instance in the superclass **must** belong to at least one subclass.
  - **Example:** Every `SHAPE` must be classified as either a `CIRCLE` , `RECTANGLE` , or `TRIANGLE` —there are no generic shapes stored in the system.
- **Partial:** Represented visually by a **single line**. An instance in the superclass **may** belong to a subclass, but it is not required to.
  - **Example:** Not every `EMPLOYEE` is a `MANAGER` ; most employees exist simply as general employees without any subclass membership.

7 Solved Exercises

📝 EXERCISE 1:

Design a comprehensive Entity-Relationship (ER) Diagram for an academic task-tracking system based strictly on the following business rules and operational constraints:

1. **Teachers:** The system tracks **Teachers**. Each teacher is uniquely identified by a `Teacher_ID` and has a `Name` and a unique `Email` address.

2. **Tasks:** Teachers create **Tasks**. However, task numbering is not unique; multiple teachers can independently create their own "Task 1", "Task 2", etc. Therefore, a task cannot be identified without knowing which teacher created it. The system tracks a sequential `Task_Number` (which resets per teacher) and a `Title`. Every task in the system must belong to exactly one teacher.
3. **Groups:** Teachers form and manage student **Groups**. Group numbers are localized to the teacher, meaning Teacher 1 can have a "Group 1" and Teacher 2 can also have a completely separate "Group 1". A group requires a `Group_No` and a custom `Group_Name`. A group cannot exist without its managing teacher.
4. **Students: Students** are independent strong entities tracked globally in the system by a unique `Student_ID`, a `Name`, and an `Email`.
5. **Team Memberships:** A student can belong to **more than one group** simultaneously. Similarly, each group can contain **multiple students**.
6. **Task Submissions & Grading:** Groups collaborate to complete assignments. A group can solve **multiple tasks**, and a single task is distributed to be solved by **multiple groups**.
7. **Operational Constraints:**
   - When a group completes a task, a shared **Mark** is issued for that specific group's submission.
   - **Every task created must be distributed to at least one group** to be solved (Total/Mandatory participation for tasks).

✅ Show Solution

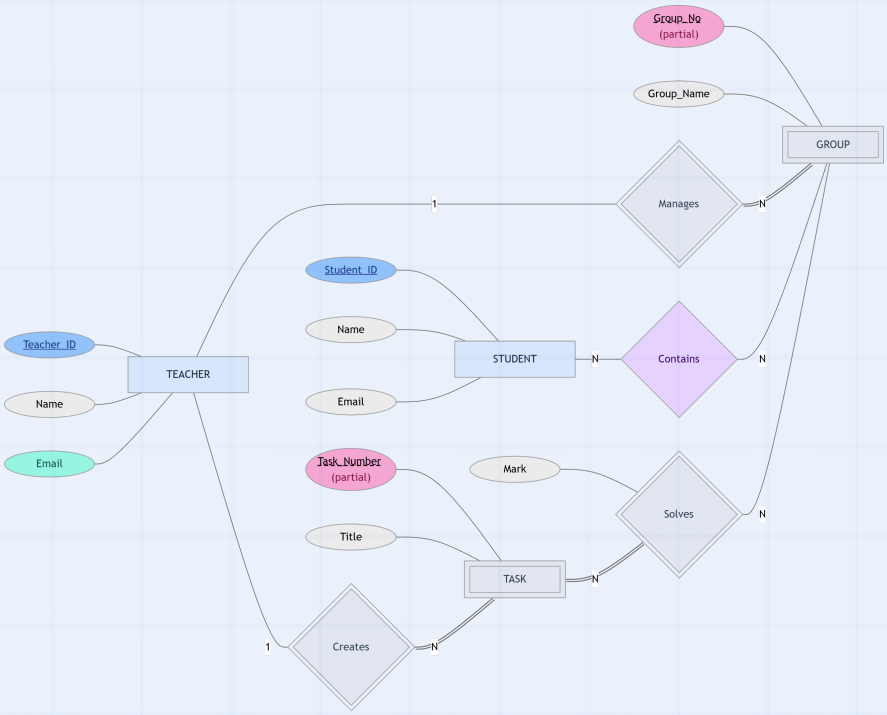


Diagram 45: academic task-tracking system